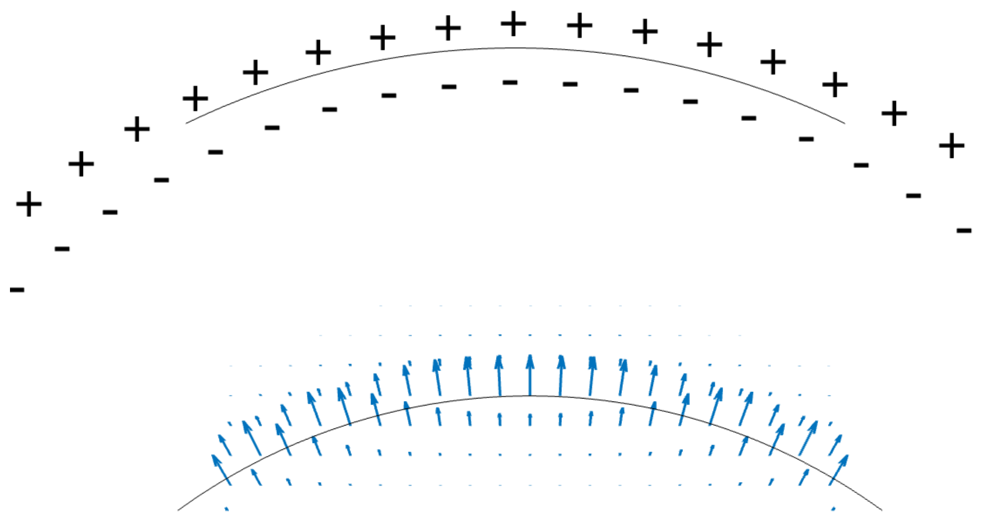
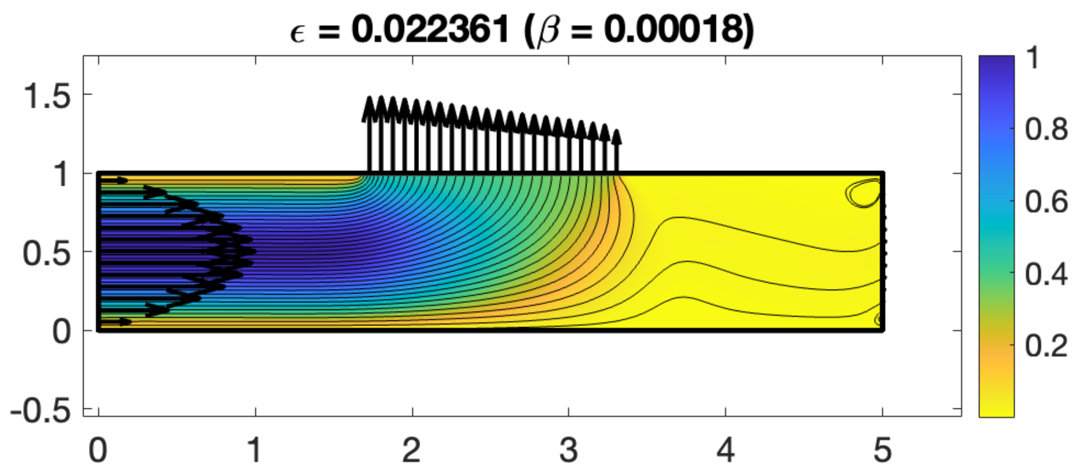
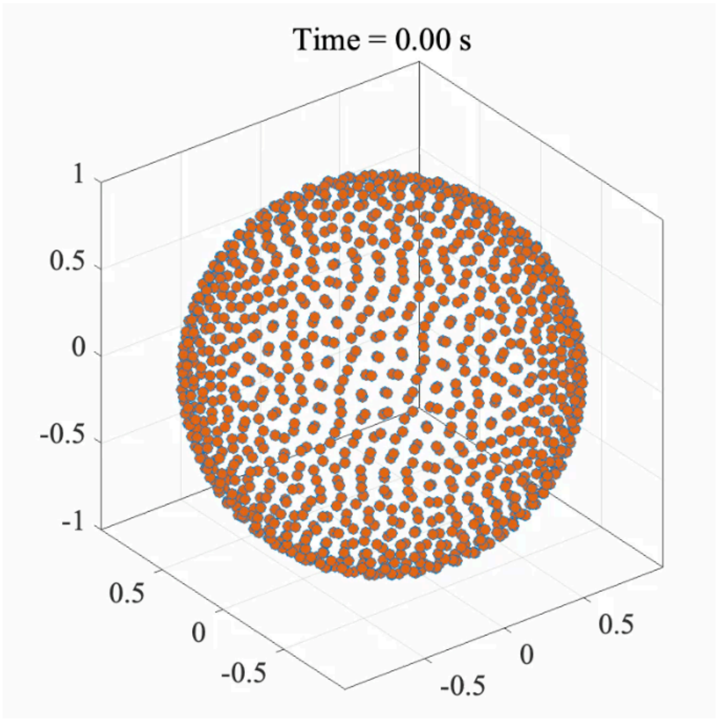
Flows Through Permeable Membranes Using the Method of Regularized Stokeslets







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ICERM-SIAM Empowering a Diverse Computational Mathematics Research Community Showcase
Third Joint SIAM/CAIMS Annual Meetings (AN25)
Friday, August 1, 2025

The Microswimmers

ICERM-SIAM Workshop Collaboration Summer 2024

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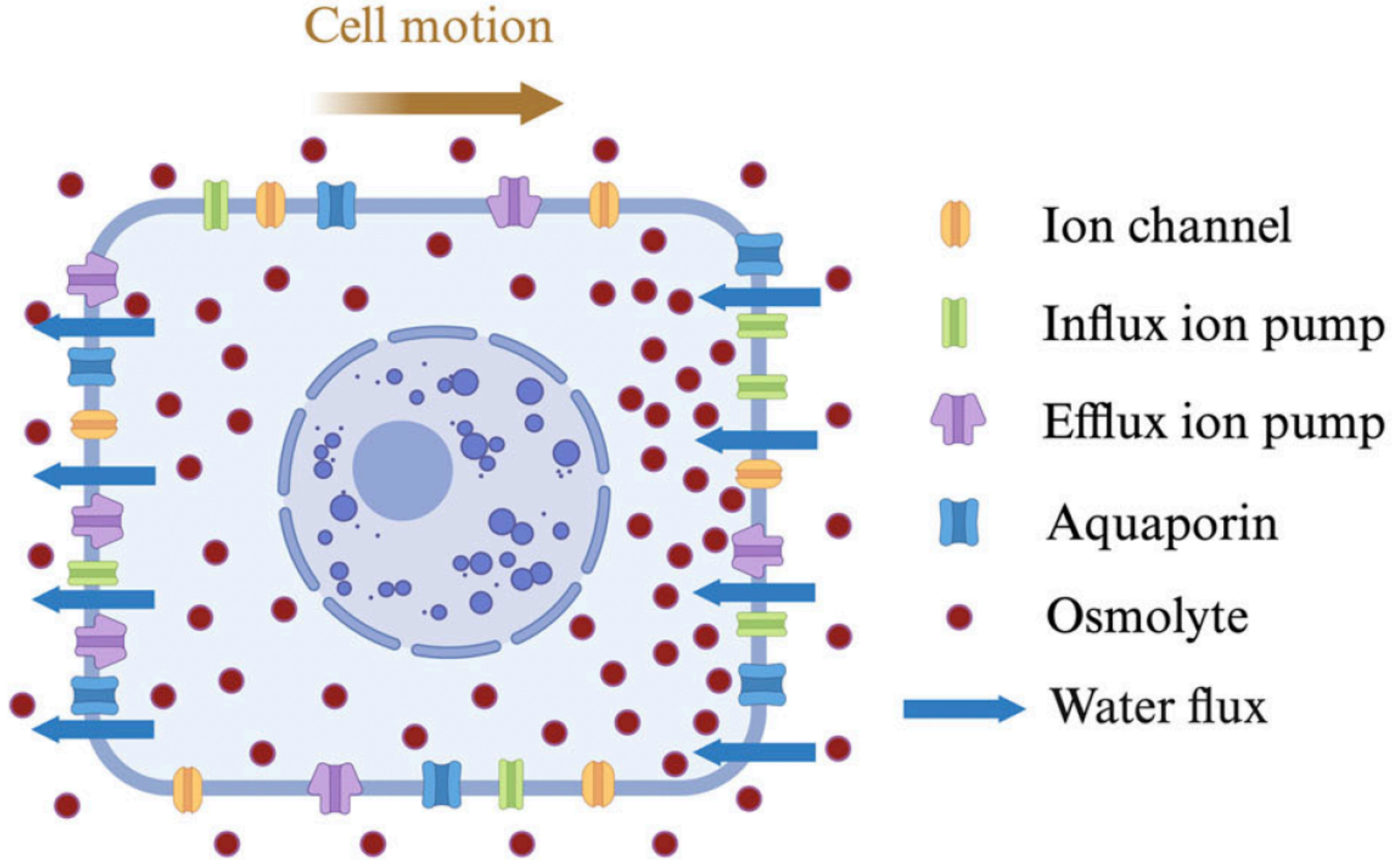
Motivation

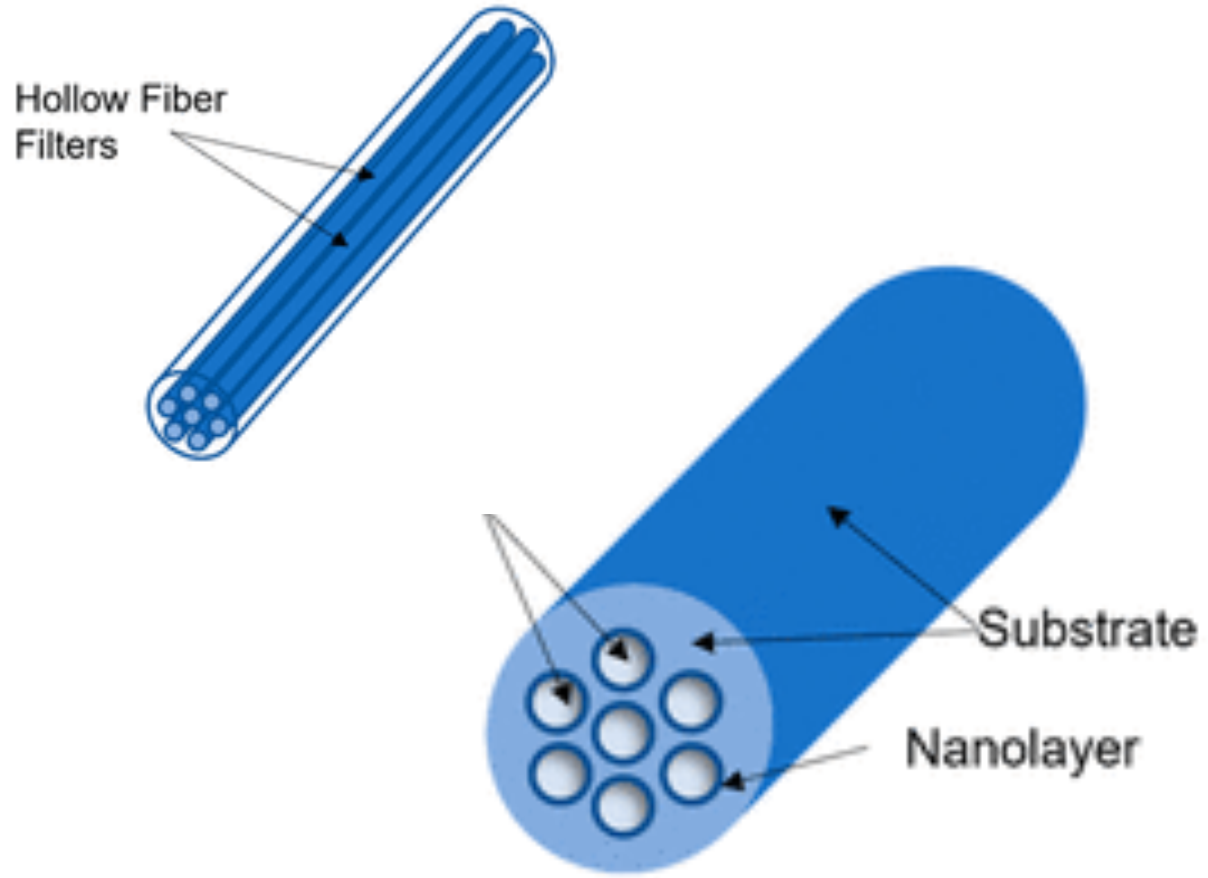
Osmotic Water Flow in Migrating Cells



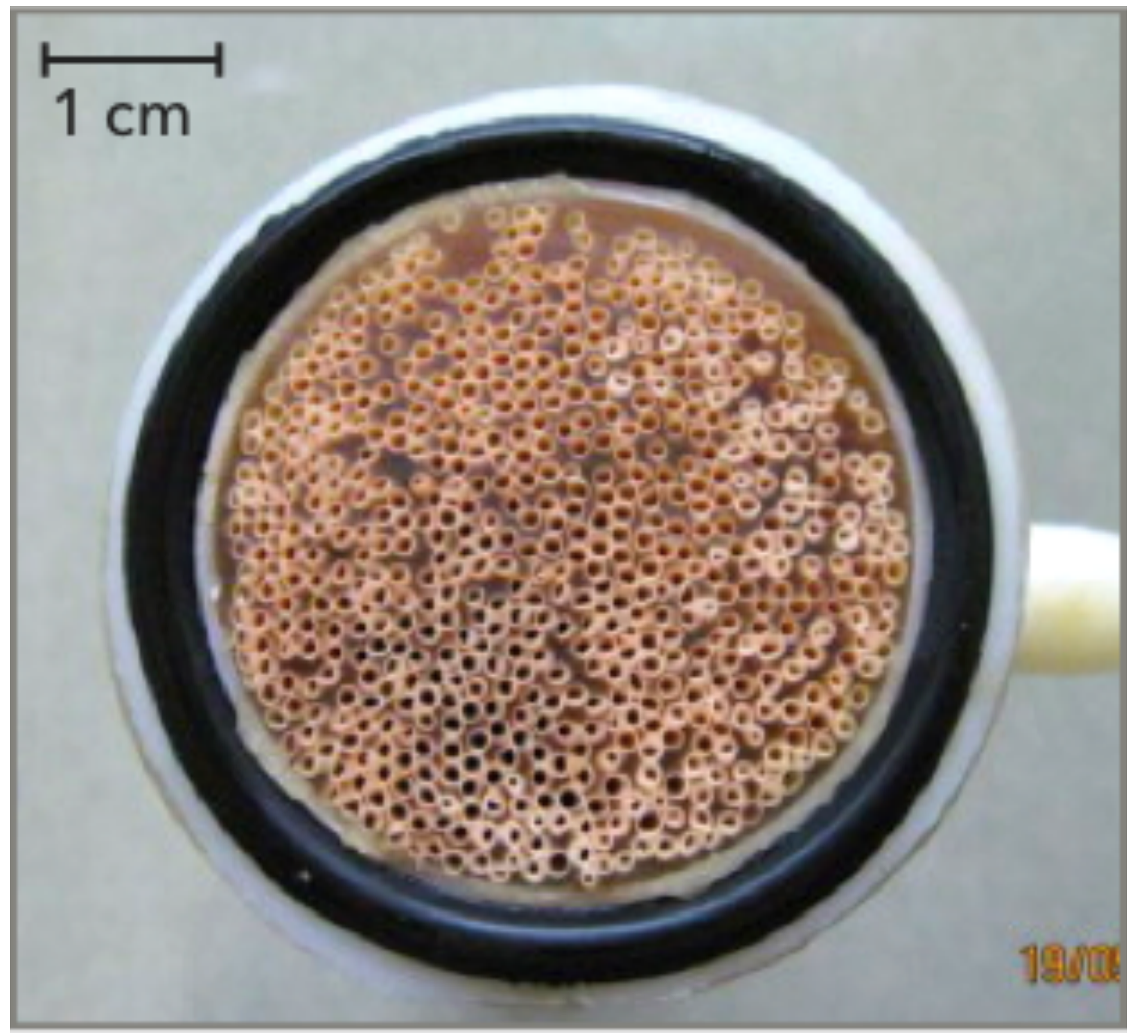
Alonso-Matilla, R., Provenzano, P.P. & Odde, D.J. Physical principles and mechanisms of cell migration. *npj Biol. Phys. Mech.* **2**, 2 (2025).

- Cells in confined geometries can be driven by water permeation (osmosis)
- Cell area and shape changes due to ion channel regulation
- Simulations using grid-based methods are computationally expensive.

Tangential Flow Filtration



Agrawal, P. et al. A Review of Tangential Flow Filtration: Process Development and Applications in the Pharmaceutical Industry, Organic Process Research & Development. Vol 27, Issue 4 2023.



L-Z Zhang, Heat and mass transfer in a randomly packed hollow fiber membrane module: A fractal model approach, International Journal of Heat and Mass Transfer, Volume 54, Issues 13–14, 2011.

- Used in agriculture, wastewater treatment, and pharmaceutical industries
- Pressure-driven flow drives separation in tangential flow filtration to reduce accumulation at the membrane

Microscale Flows Driven by External Forcing

Reynolds number:

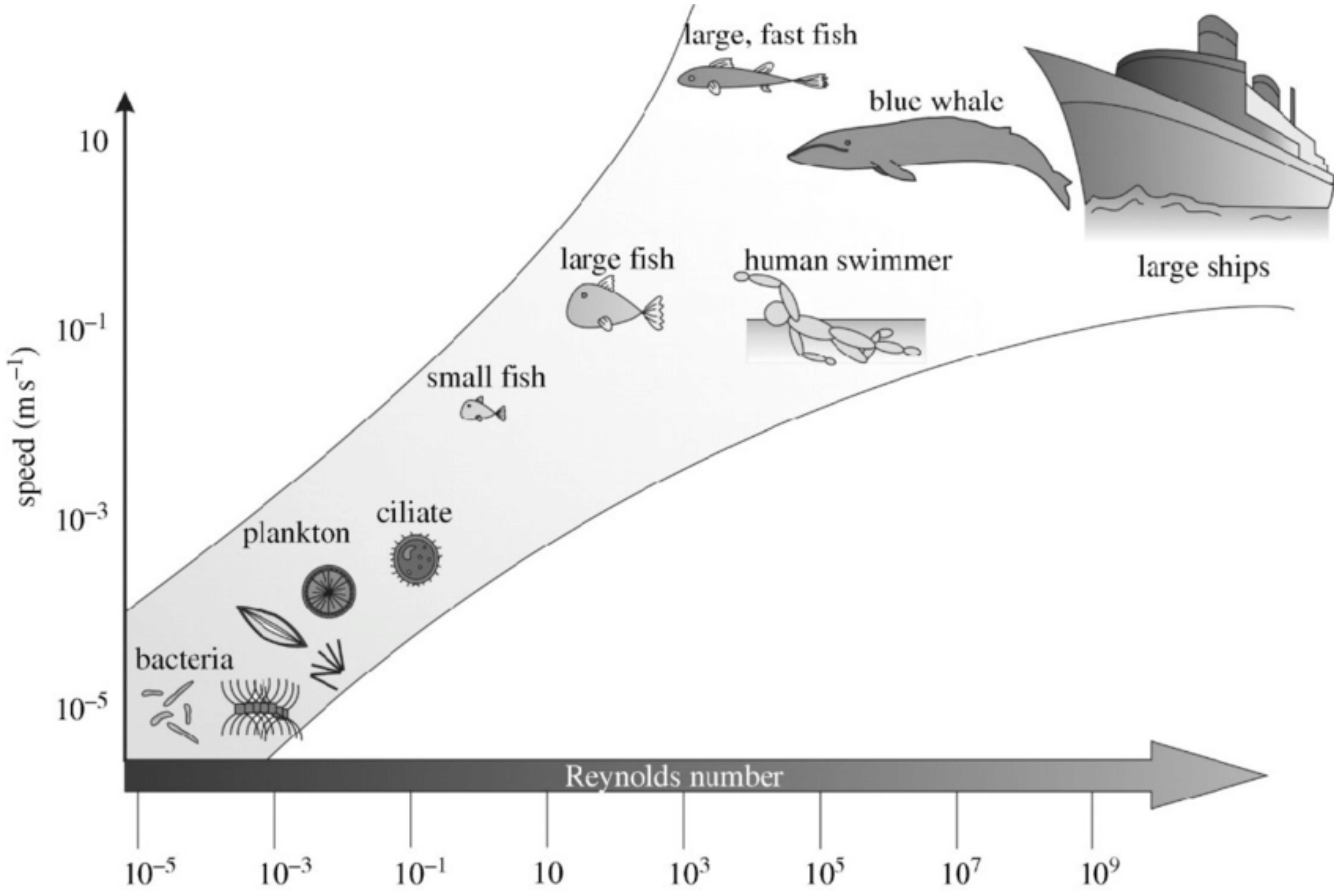
$$\text{Re} = \frac{\rho U L}{\mu} = \frac{\text{(density)(speed)(length)}}{\text{(viscosity)}}$$

<u>Stokes Equations:</u> (Re << 1, microorganisms, slow-moving particles, very viscous flows)

$$\nabla p - \mu \nabla^2 \mathbf{u} = \mathbf{F}, \quad \nabla \cdot \mathbf{u} = 0$$

$\mathbf{u} =$ fluid velocity, $p =$ fluid pressure,

$\mu =$ viscosity, $\mathbf{F} =$ force

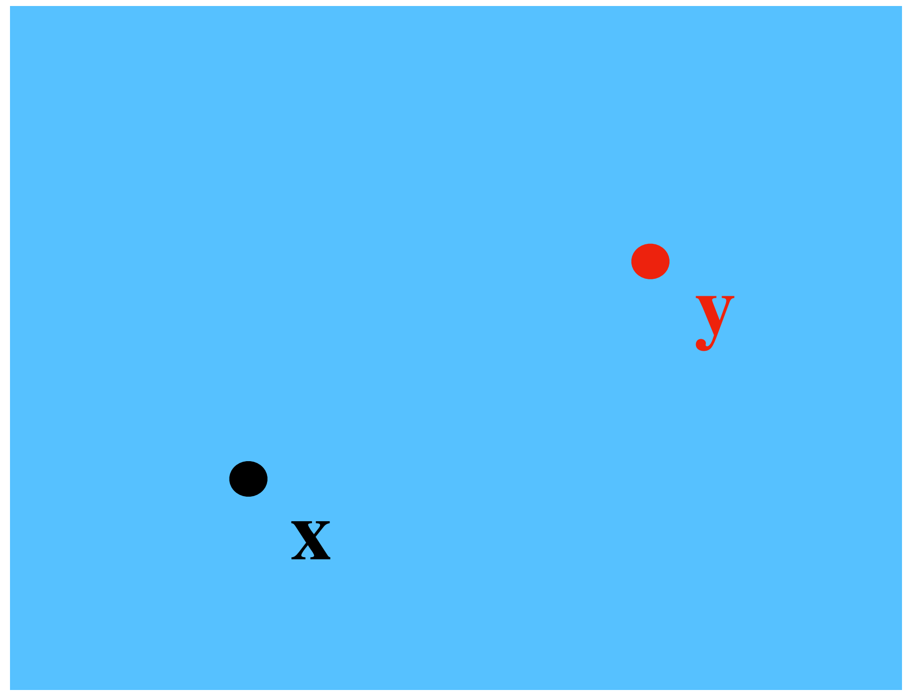


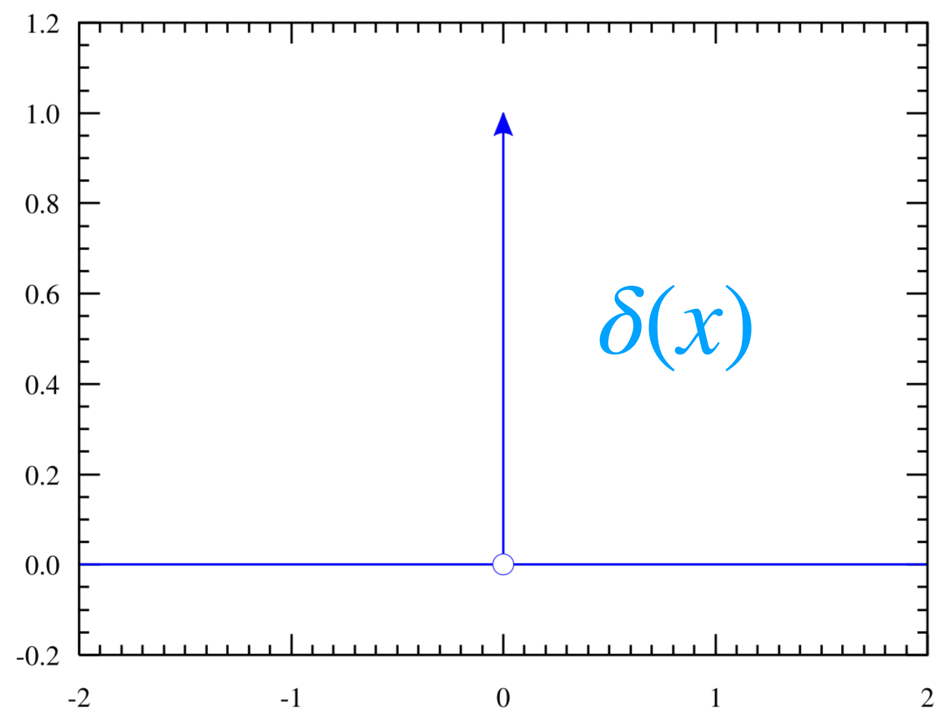
M. Salta et al., "Designing biomimetic antifouling surfaces," Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, vol. 368, no. 1929, pp. 4729–4754, Oct. 2010.

The Point-Force Problem and the Stokeslet (St)

Stokes Flow at $\mathbf{x}$ driven by a **point-force** at $\mathbf{y}$:

$$\nabla p - \mu \nabla^2 \mathbf{u} = \mathbf{f}\delta(\mathbf{x}-\mathbf{y}), \quad \nabla \cdot \mathbf{u} = 0$$





has a **fundamental solution (Stokeslet)**

$$p(\mathbf{x}) = \mathbf{f}(\mathbf{y}) \cdot \nabla G(\mathbf{x}-\mathbf{y})$$

$$\mu \mathbf{u}(\mathbf{x}) = \big(\mathbf{f}(\mathbf{y}) \cdot \nabla\big) \nabla B(\mathbf{x}-\mathbf{y}) - \mathbf{f}(\mathbf{y}) G(\mathbf{x}-\mathbf{y})$$

where $G(\mathbf{x})$ is the solution to $\nabla^2 G = \delta$ and $B(\mathbf{x})$ is solution to $\nabla^2 B = G$

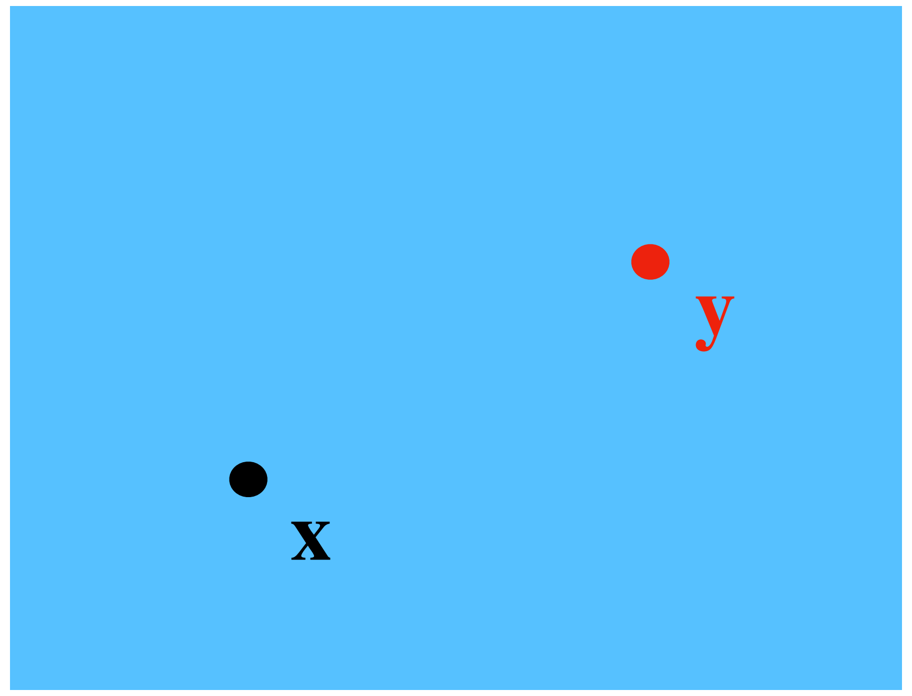
- **2D:** $G(\mathbf{x}-\mathbf{y}) = \dfrac{\ln(r)}{2\pi}, \quad B(\mathbf{x}-\mathbf{y}) = \dfrac{r^2}{8\pi}(\ln(r) - 1)$
- **3D:** $G(\mathbf{x}-\mathbf{y}) = -\dfrac{1}{4\pi r}, \quad B(\mathbf{x}-\mathbf{y}) = -\dfrac{r}{8\pi}$

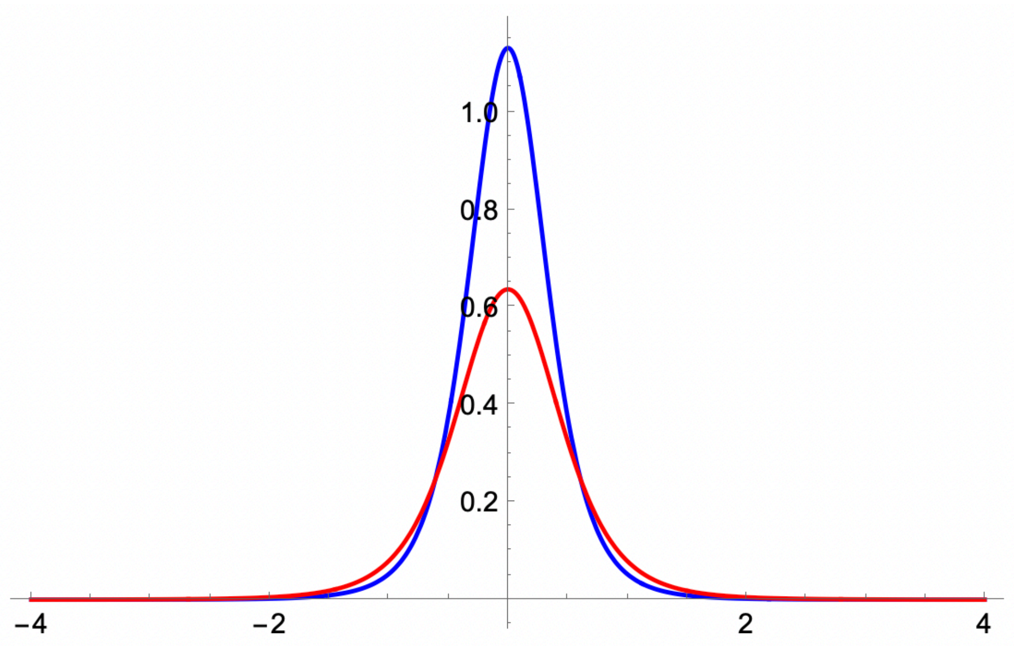
Here we set
$r = \|\mathbf{x}-\mathbf{y}\|$

The *Regularized* Stokeslet Problem and Solution

Use *smooth* approximation to $\delta(\mathbf{x}) \approx \phi_\epsilon(\mathbf{x})$ (“blob”)

$$\nabla p - \mu \nabla^2 \mathbf{u} = \mathbf{f}\, \phi_\epsilon(\mathbf{x} - \mathbf{y})$$
$$\nabla \cdot \mathbf{u} = 0$$





Sample ϕ_ϵ “blob” functions in 1D and 2D

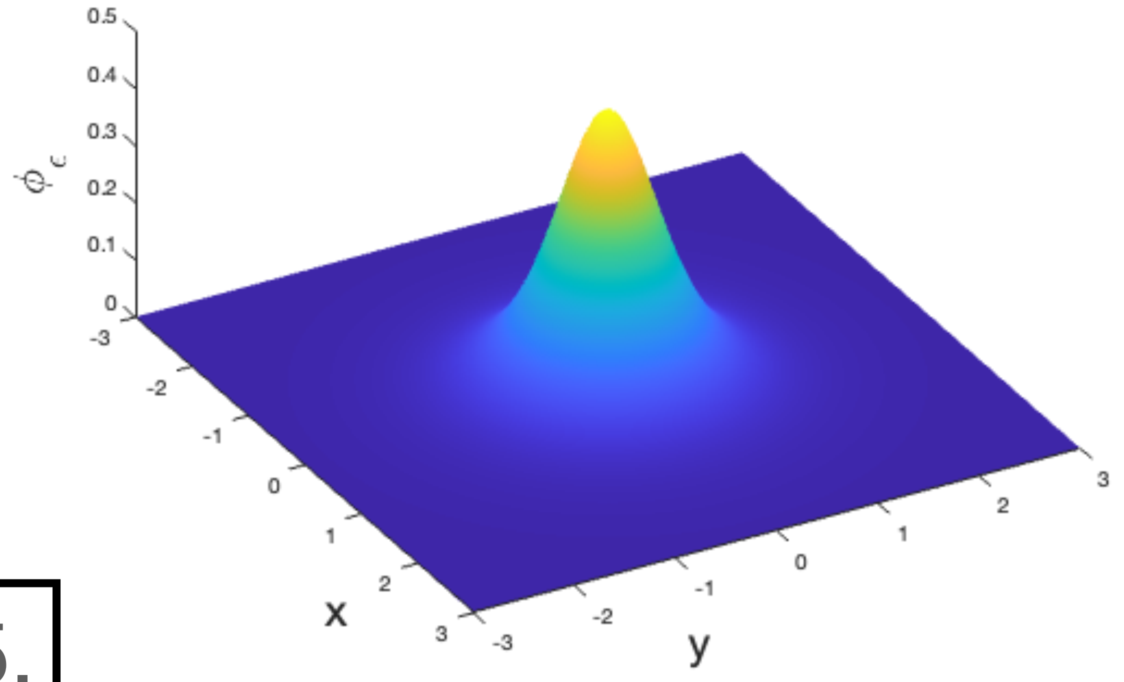
This has an analytical solution called the ***regularized Stokeslet***[1]:

$$p(\mathbf{x}) = \mathbf{f}(\mathbf{y}) \cdot \nabla G_\epsilon(\mathbf{x} - \mathbf{y})$$

$$\mu \mathbf{u}(\mathbf{x}) = \big(\mathbf{f}(\mathbf{y}) \cdot \nabla\big) \nabla B_\epsilon(\mathbf{x} - \mathbf{y}) - \mathbf{f}(\mathbf{y}) G_\epsilon(\mathbf{x} - \mathbf{y})$$

where and $\nabla^2 G_\epsilon = \phi_\epsilon$, and $\nabla^2 B_\epsilon = G_\epsilon$.

$$\int_\Omega \phi_\epsilon \, d\mathbf{x} = 1, \lim_{\epsilon \to 0} \phi_\epsilon = \delta$$

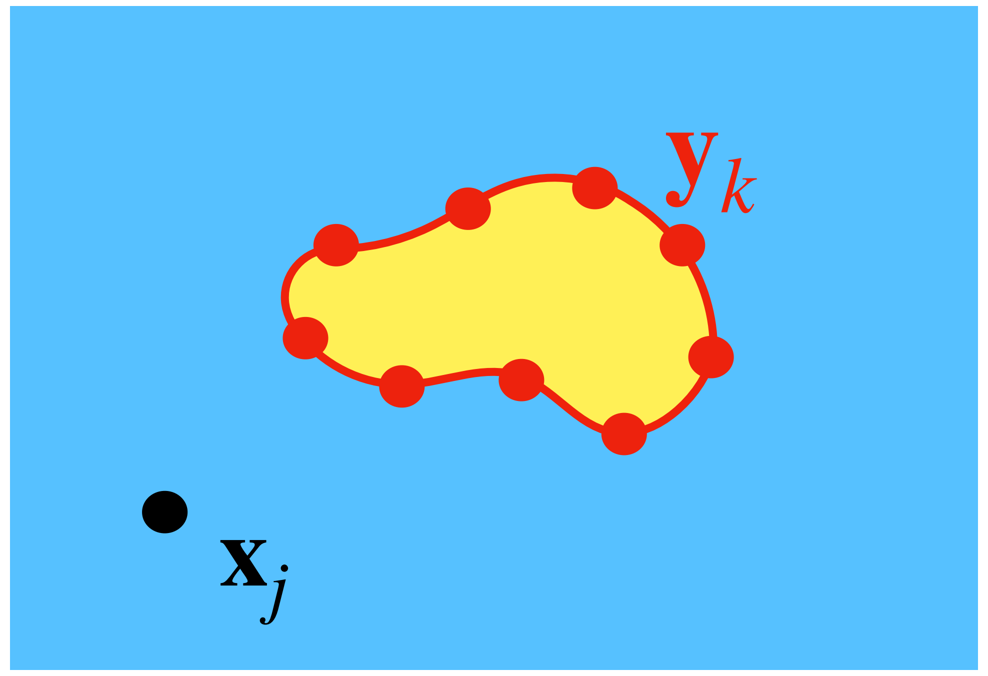


1. R. Cortez, The method of regularized Stokeslets, SIAM J. Sci. Comput. 23 (4) (2001) 1204–1225.

Method of Regularized Stokeslets (MRS)

(Cortez, 2001) Discretize surface and sum over *smoothed* point-forces

$$\nabla p - \mu \nabla^2 \mathbf{u} = \sum_{k=1}^{N} \mathbf{f}_k \phi_\epsilon(\mathbf{x}_j - \mathbf{y}_k), \quad \nabla \cdot \mathbf{u} = 0$$

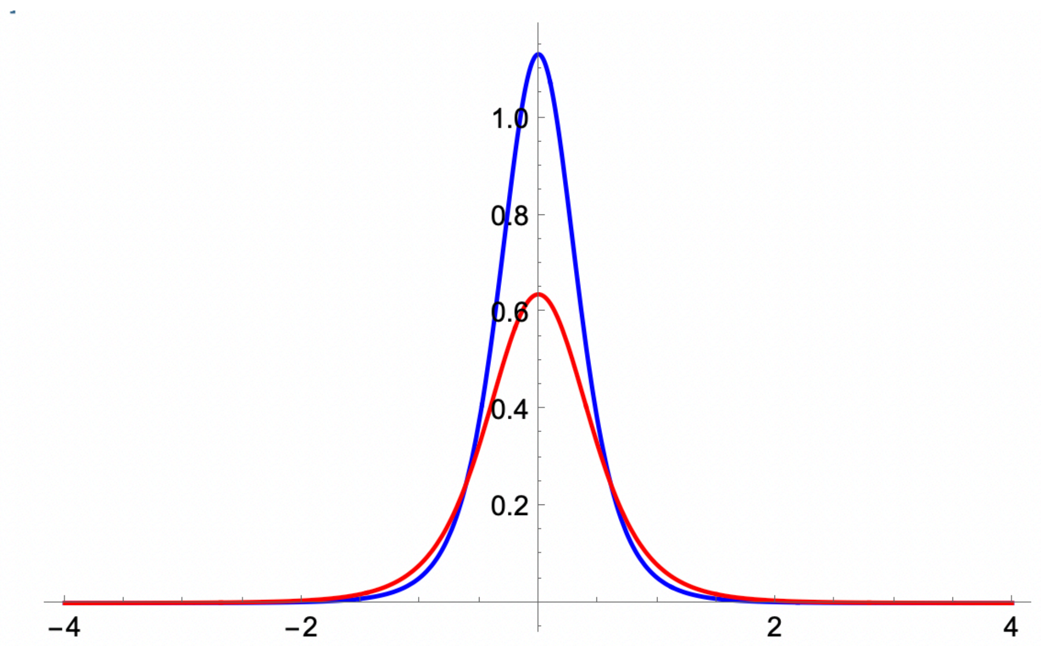


Regularized Stokeslet solution given N smoothed point forces is

$$p(\mathbf{x}_j) = \sum_{k=1}^{N} \mathbf{f}_k \cdot \nabla G_\epsilon(\mathbf{x}_j - \mathbf{y}_k)$$

$$\mathbf{u}(\mathbf{x}_j) = \frac{1}{\mu} \sum_{k=1}^{N} \Big((\mathbf{f}_k \cdot \nabla) \nabla B_\epsilon(\mathbf{x}_j - \mathbf{y}_k) - \mathbf{f}_k G_\epsilon(\mathbf{x}_j - \mathbf{y}_k) \Big)$$

Every $\mathbf{y}_k$ source point has a “blob” force



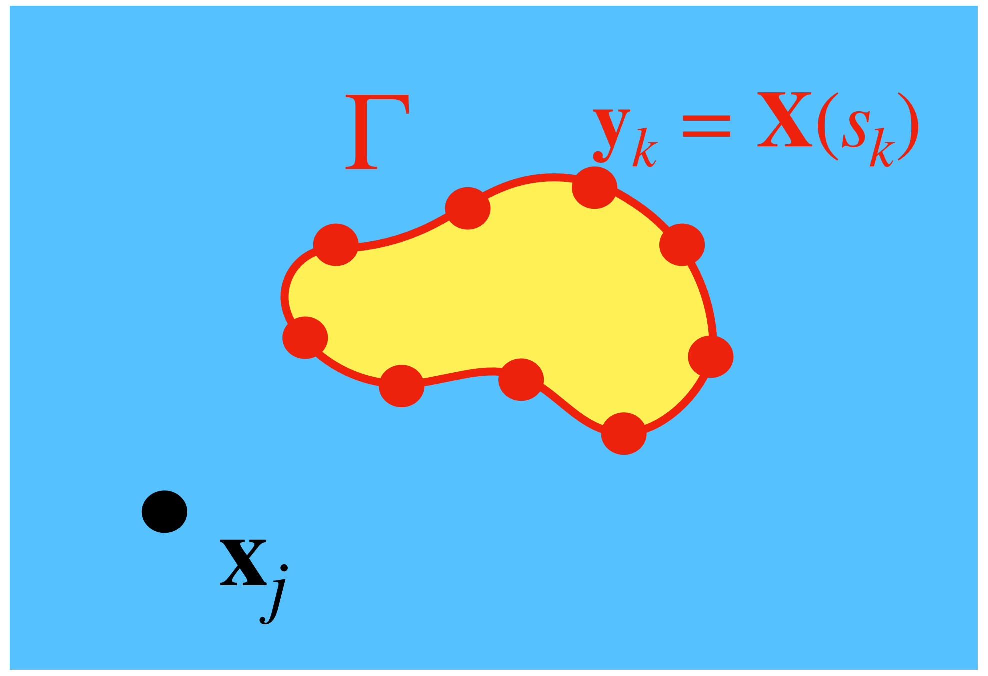
Given $j = 1,2,\ldots,M$ target points $\longrightarrow 2M \times 2N$ linear system (in 2D):

$$\overrightarrow{\mathscr{U}} = \mathscr{M}_{St} \overrightarrow{\mathscr{F}}$$

Permeable (moving) Boundary Problem

Consider a permeable (moving) boundary Γ, with parametrization $\mathbf{X}(s)$.

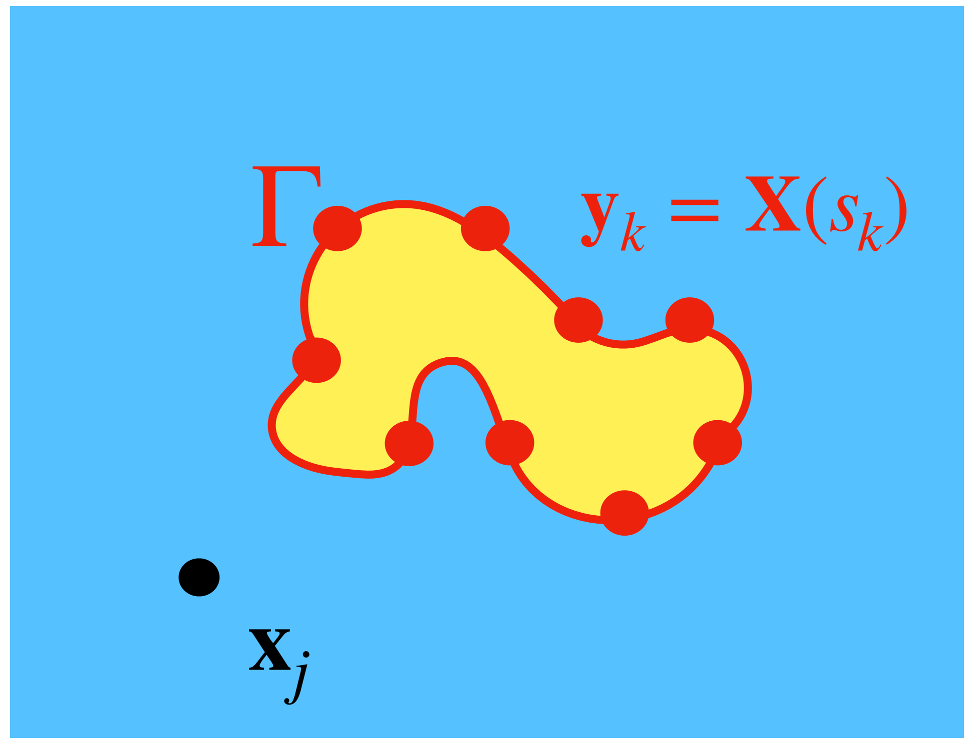
Boundary Γ at time t_1



- $\mathbf{u}(\mathbf{x})$ = fluid particle velocity
- $\mathbf{U}$ = difference between membrane and fluid velocity
- Fluid Domain velocity field (including points on membrane):

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}(\mathbf{x}) + \boxed{\int_\Gamma \mathbf{U}(s)\phi_\epsilon(\mathbf{x} - \mathbf{X}(s))ds}$$

Boundary Γ at time t_2



- Regularized Model Equations (in integral form)[2]:

$$\nabla p - \mu \nabla^2 \mathbf{u} = \int_\Gamma \mathbf{f}(s)\phi_\epsilon(\mathbf{x} - \mathbf{X}(s))ds, \quad \nabla \cdot \mathbf{u} = \int_\Gamma \mathbf{U} \cdot \nabla \phi_\epsilon(\mathbf{x} - \mathbf{X}(s))ds$$

2. R. Cortez, M. Hernandez-Viera, O. Richfield, A model for Stokes flow in domains with permeable boundaries, Fluids 6 (11) (2021).

Solution to the Regularized Problem

Split into 2 subproblems:

1. Regularized Force, Zero Divergence:

$$\nabla p - \mu \nabla^2 \mathbf{u} = \int_\Gamma \mathbf{f}(s)\phi_{\epsilon_1}(\mathbf{x} - \mathbf{X}(s))ds, \quad \nabla \cdot \mathbf{u} = 0$$

Regularized Stokeslet Problem

$$\overrightarrow{\mathscr{U}} = \mathscr{M}_{St}\overrightarrow{\mathscr{F}}$$

2. No Forcing, Nonzero Regularized Divergence:

$$\nabla q - \mu \nabla^2 \mathbf{v} = \mathbf{0}, \quad \nabla \cdot \mathbf{v} = -\int_\Gamma \frac{\beta(\mathbf{f}\cdot\hat{\mathbf{n}})}{\mu}\hat{n}(s) \cdot \nabla \phi_{\epsilon_2}(\mathbf{x} - \mathbf{X}(s))ds$$

Regularized *Source Doublet* Problem

$$\overrightarrow{\mathscr{V}} = \mathscr{M}_{SD}\overrightarrow{\mathscr{F}}$$

Replace $\mathbf{U}$ with $\beta(\mathbf{f}\cdot\mathbf{n})$, where $\beta = \beta(s)$ relates to permeability of Γ

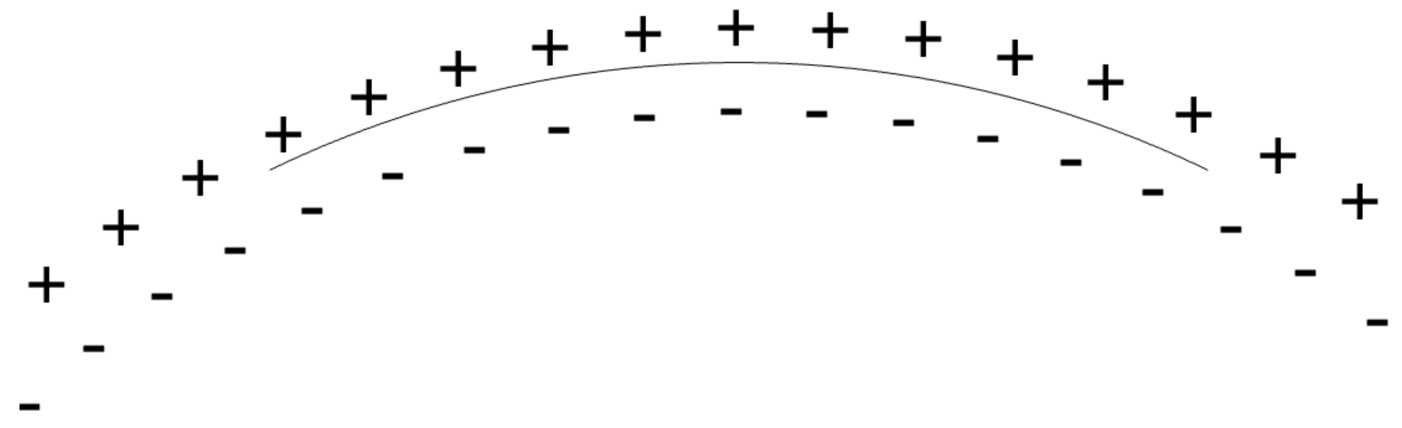
Solution: Fluid particle velocity $\mathbf{u}(\mathbf{x}) = d\mathbf{x}/dt$, membrane velocity $d\mathbf{X}/dt = \mathbf{u}(\mathbf{X}) + \mathbf{v}(\mathbf{X})$

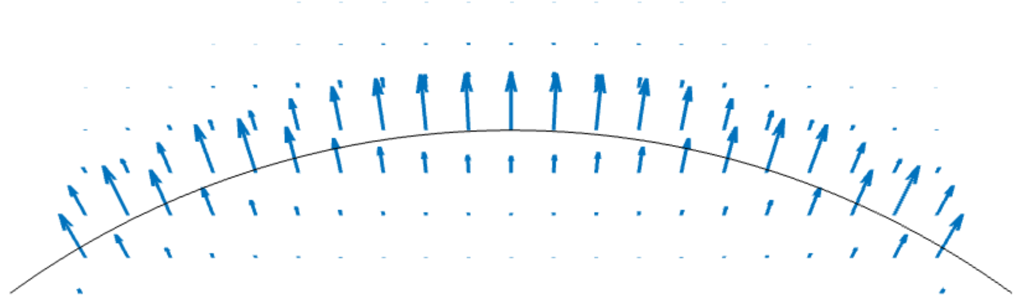
The Source Doublet (SD) and Permeability

Idea: flux across membrane is due to distribution of source/sink pairs along boundary.

$$\nabla q - \mu \nabla^2 \mathbf{v} = \mathbf{0}, \quad \nabla \cdot \mathbf{v} = - \int_\Gamma \frac{\beta(\mathbf{f} \cdot \hat{\mathbf{n}})}{\mu} \hat{n}(s) \cdot \nabla \phi_\epsilon(\mathbf{x} - \mathbf{X}(s)) ds$$

where $\beta = \beta(s)$ relates to permeability of Γ.





Solution: regularized source doublet pressure and velocity

$$q(\mathbf{x}) = - \int_\Gamma \beta(\mathbf{f} \cdot \hat{\mathbf{n}})(\hat{\mathbf{n}} \cdot \nabla \phi_\epsilon(\mathbf{x} - \mathbf{X}(s))\, ds,$$

$$\mu \mathbf{v}(\mathbf{x}) = - \int_\Gamma \beta(\mathbf{f} \cdot \hat{\mathbf{n}})(\hat{n} \cdot \nabla)(\nabla G_\epsilon(\mathbf{x} - \mathbf{X}(s)))\, ds$$

$k = 1,2,\ldots,N$ source doublet points ($\mathbf{X}(s_k)$)

$$\xrightarrow[]{}$$

$j = 1,2,\ldots,M$ target points $\mathbf{x}_j$

$$\vec{\mathscr{V}} = \mathscr{M}_{SD} \vec{\mathscr{F}}$$

Problem 1: 3D Spherical Membrane Under Tension

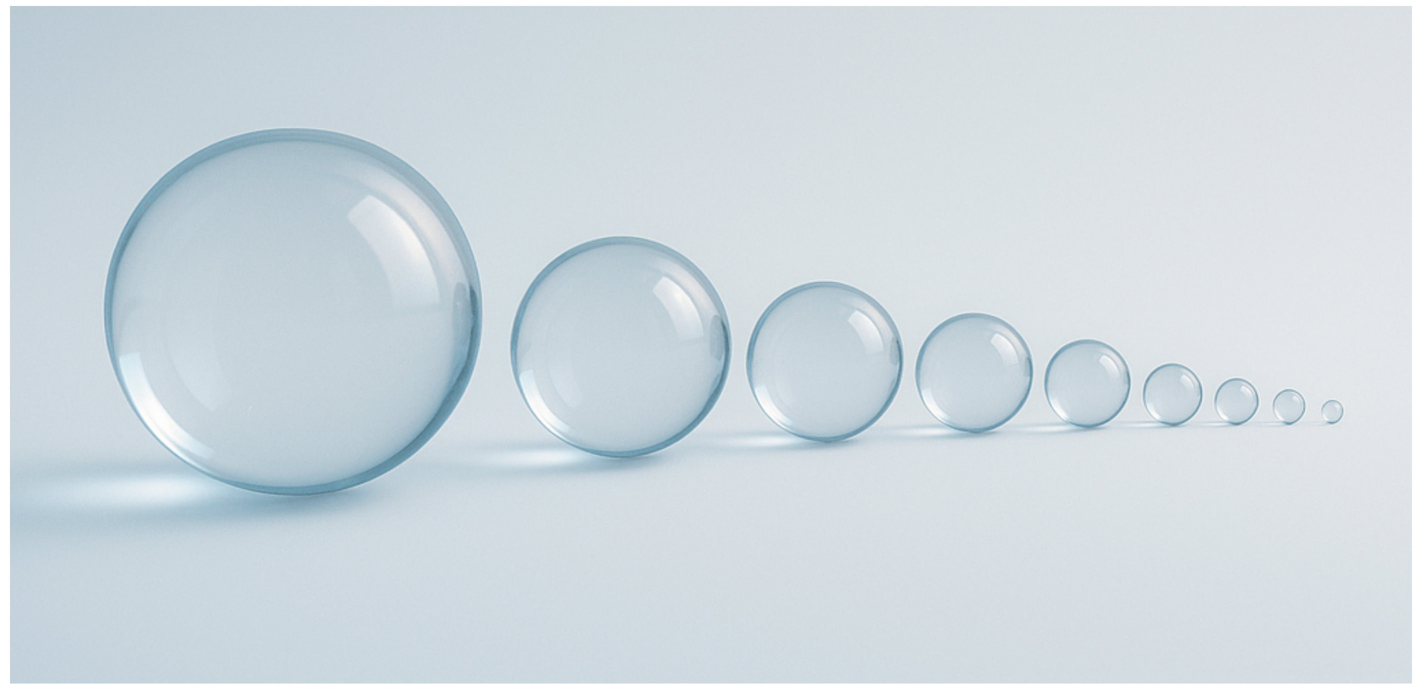
Model Validation and β Calibration – Wanda Strychalski

- Permeable, infinitely thin elastic membrane, experiencing forces due to surface tension, shrinking to a point.

- From Laplace's law: $[[p]] = \mathbf{f}(s) \cdot \hat{\mathbf{n}} = 2\kappa$

  $\Longrightarrow$ membrane velocity is $\mathbf{U} = -\dfrac{\alpha}{\mu}[[p]]\hat{\mathbf{n}} = -\dfrac{\alpha}{\mu}2\kappa\hat{\mathbf{n}}$, where

  α = permeability of membrane and κ is the curvature.



OpenAI. (2025). *Shrinking spheres illustration* [AI-generated image]. Microsoft Copilot

- Exact solution obtained by solving the ODE. ($R(t)$ = radius at time t)

$$\frac{dR}{dt} = -\frac{2\alpha}{\mu R}, \quad R(0) = R_0, \quad R_{exact}(t) = \sqrt{R_0^2 - \frac{4\alpha}{\mu}t} \text{ for } 0 \le t \le \frac{R_0^2\mu}{4\alpha}$$

Problem 1: Calibrating β

Calibrating β using the true solution

- Velocity is entirely from source doublets: $\beta(\mathbf{f}\cdot\mathbf{n}) = \beta(2\kappa)$

$$\mathbf{U}(\mathbf{x}) = \mathbf{0} + \mathbf{v}(\mathbf{x}) = -\frac{1}{\mu}\int_\Gamma \beta(2\kappa)(\hat{\mathbf{n}}\cdot\nabla)(\nabla G_\epsilon(\mathbf{x}-\mathbf{X}(s)))\,ds$$

- Choose a blob and calculate source doublet velocity, $\mathbf{v}(\mathbf{x})$.

  Select β so the velocity is consistent is the exact solution: $\boxed{\dfrac{dR}{dt} = \mathbf{v}\cdot\hat{\mathbf{n}} = -\dfrac{\alpha}{\mu}2\kappa.}$

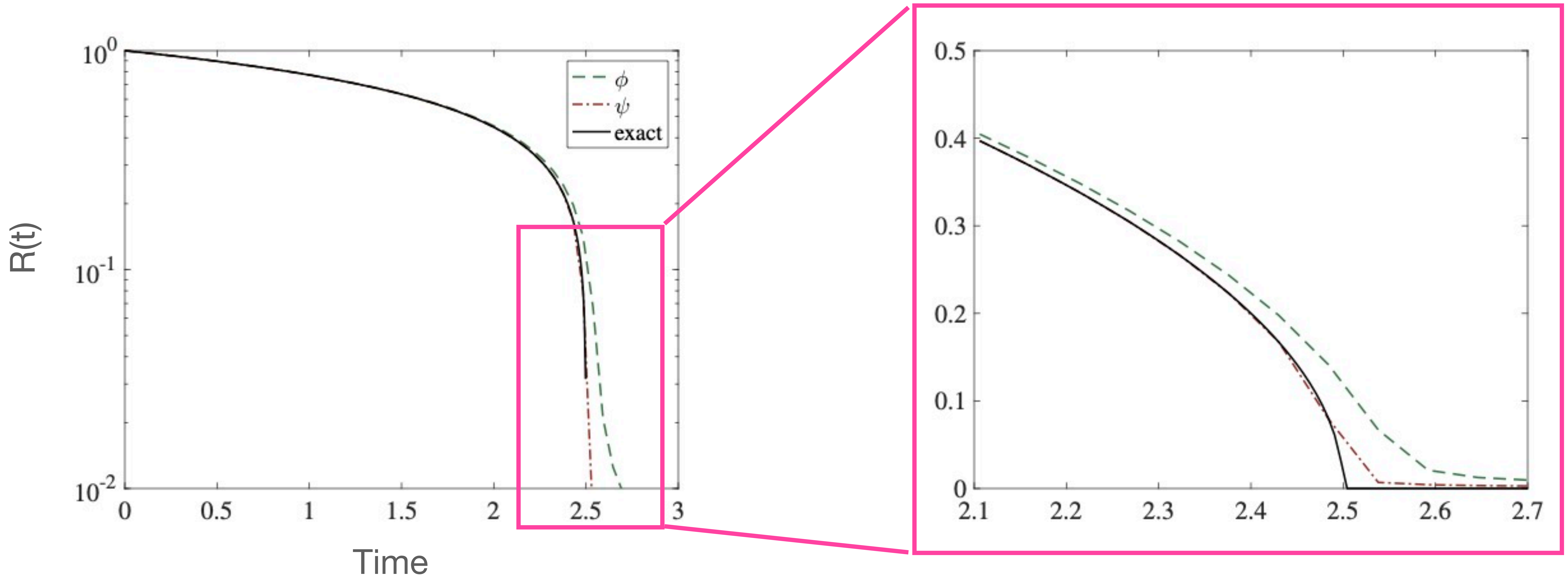
- **β depends on the blob, regularization size, and the permeability of the membrane!**

$$\phi_\epsilon(r) = \frac{15\epsilon^4}{8\pi(r^2+\epsilon^2)^{7/2}} \implies \mathbf{v}\cdot\hat{\mathbf{n}} = \boxed{-\frac{2\beta\kappa}{\epsilon\mu}\left(\frac{3}{4}\right.}\left.-\frac{\epsilon^2}{4R^2}+O\left(\frac{\epsilon^5}{R^5}\right)\right) \implies \boxed{\beta = \frac{4}{3}\epsilon\alpha}$$

$$\psi_\epsilon(r) = \frac{45\epsilon^8}{8\pi(r^2+\epsilon^2)^{11/2}}\left(\frac{10}{3}-\frac{37r^2}{6\epsilon^2}+\frac{r^4}{\epsilon^4}\right) \implies \mathbf{v}\cdot\hat{\mathbf{n}} = \boxed{-\frac{2\beta\kappa}{\epsilon\mu}\left(\frac{9}{4}\right.}\left.+\frac{21\epsilon^5}{128R^5}+O\left(\frac{\epsilon^7}{R^7}\right)\right) \implies \boxed{\beta = \frac{4}{9}\epsilon\alpha}$$

Problem 1: Results

Comparing Radius of computed to actual solution

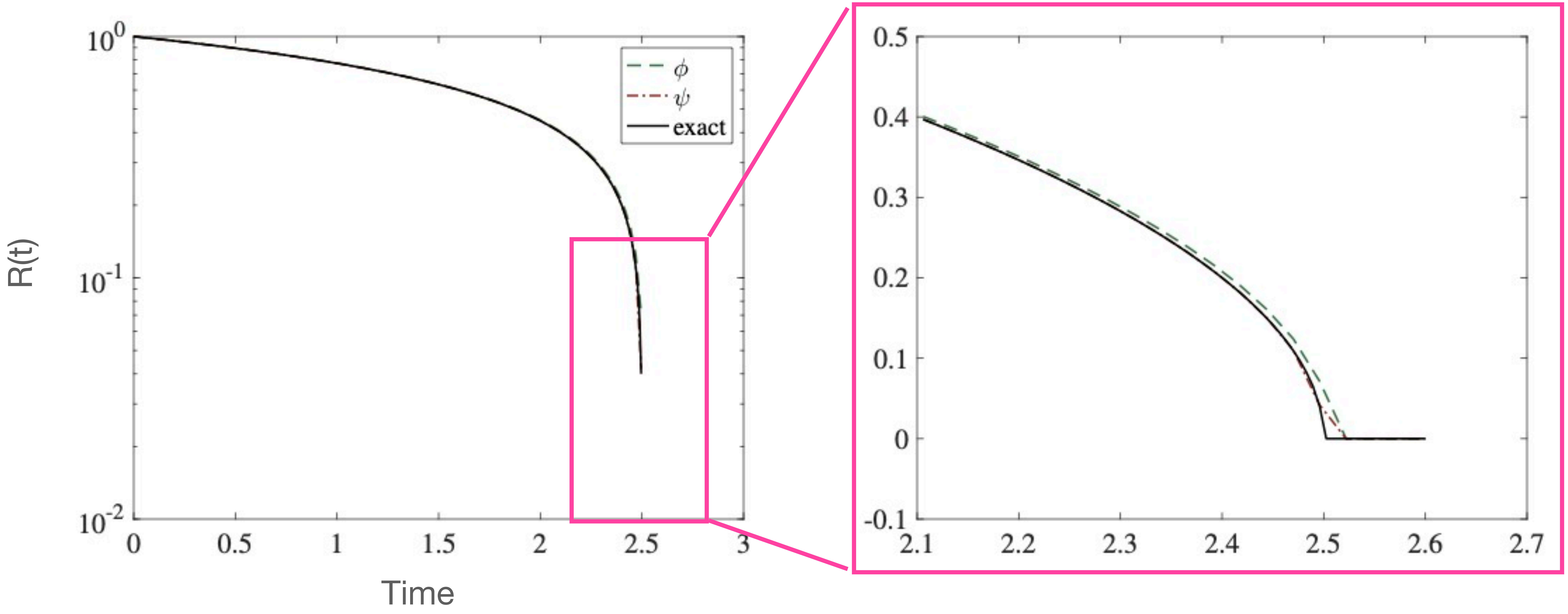


$$R(0) = 1,\ \alpha = 0.1 = \epsilon$$

$N = 1038$, computed using both blobs $\phi_\epsilon(r)$ (basic) and $\psi_\epsilon(r)$ (satisfies extra conditions for higher accuracy)

Problem 1: Results

Comparison with dynamic rescaling

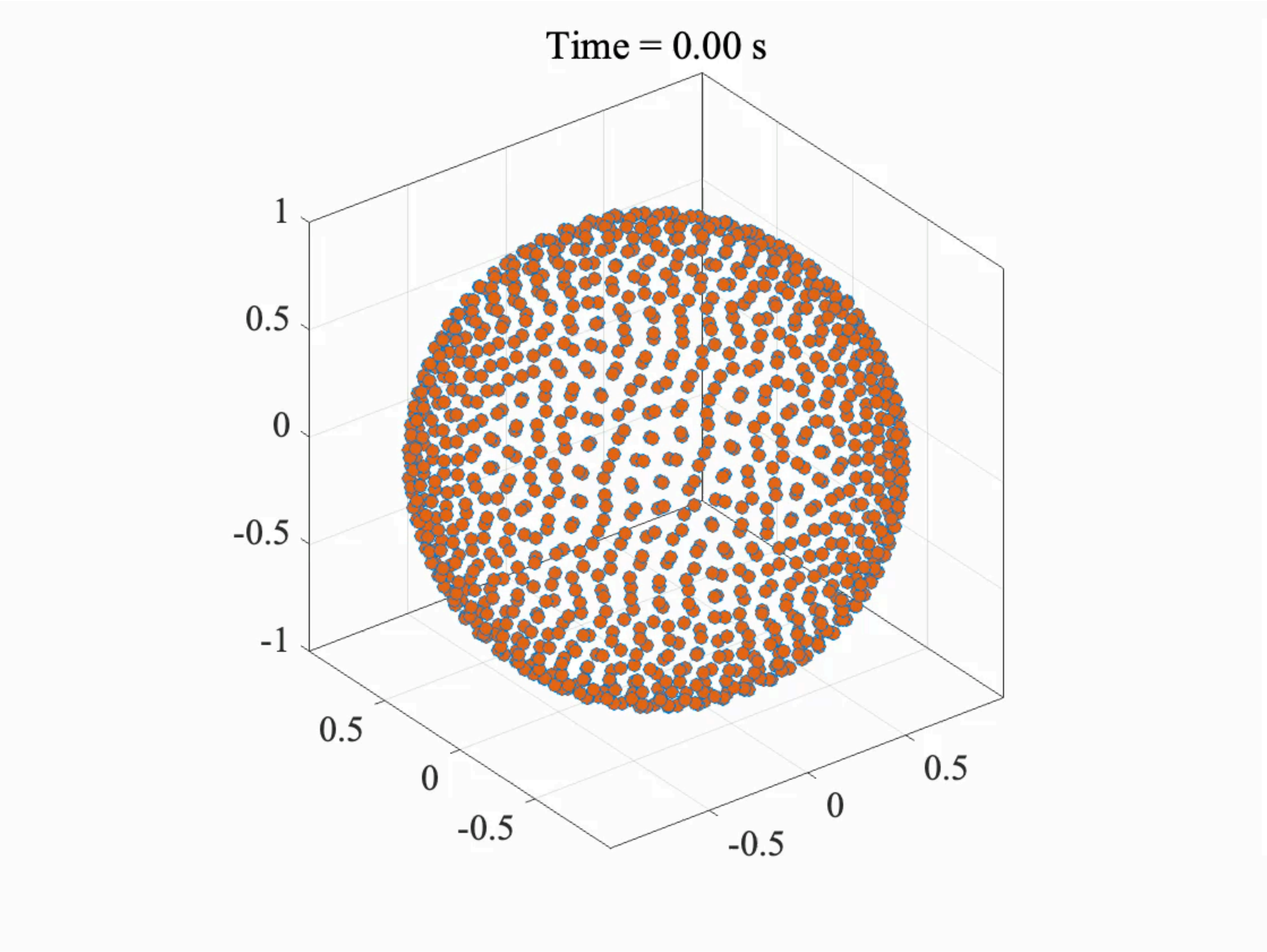


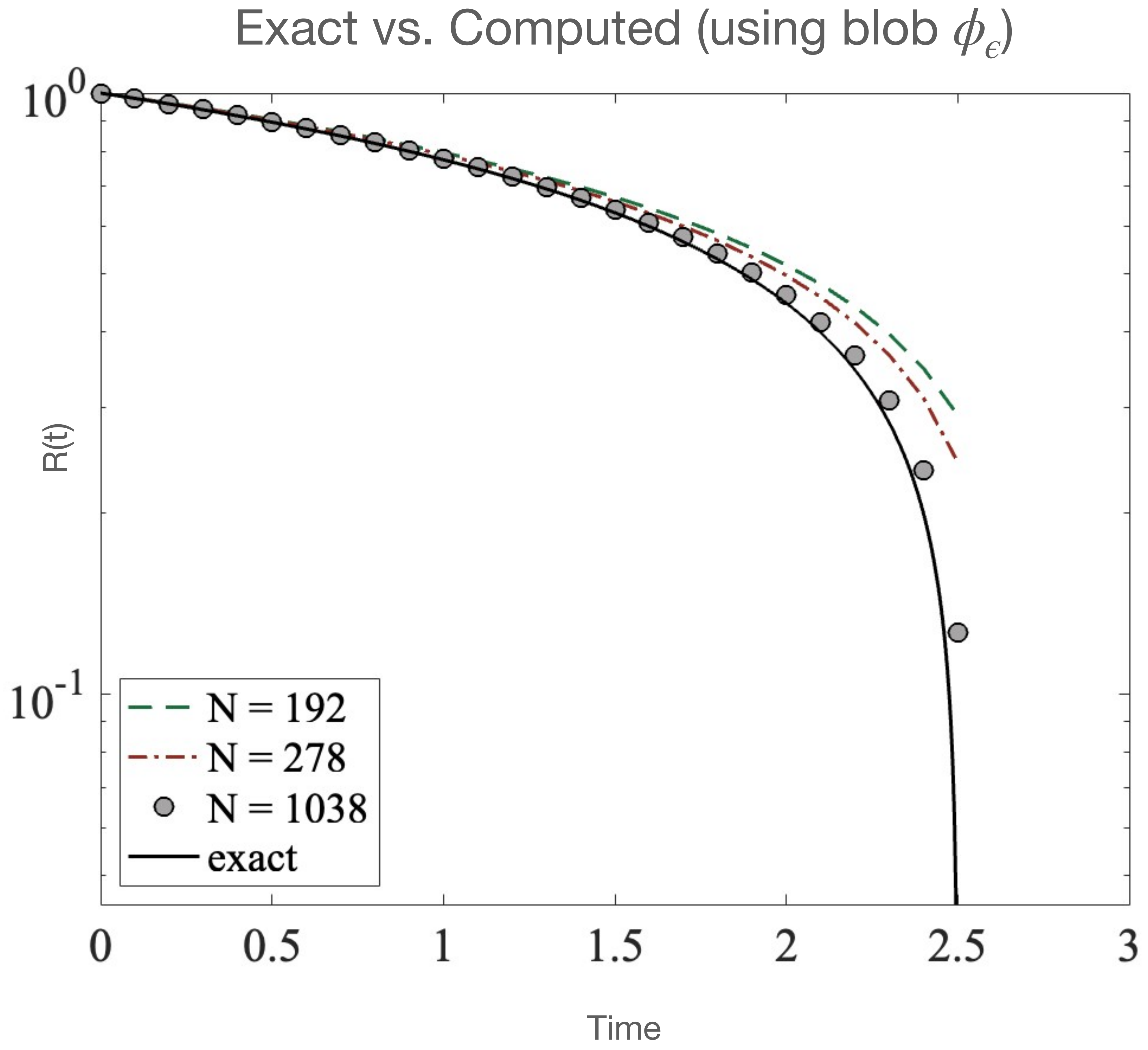
$$R(0) = 1,\ \alpha = 0.1,\ \epsilon = 0.1R(t)$$

$N = 1038$, computed using both blobs $\phi_\epsilon(r)$ (basic) and $\psi_\epsilon(r)$ (satisfies extra conditions for higher accuracy)

Problem 1: Results

Simulation and convergence



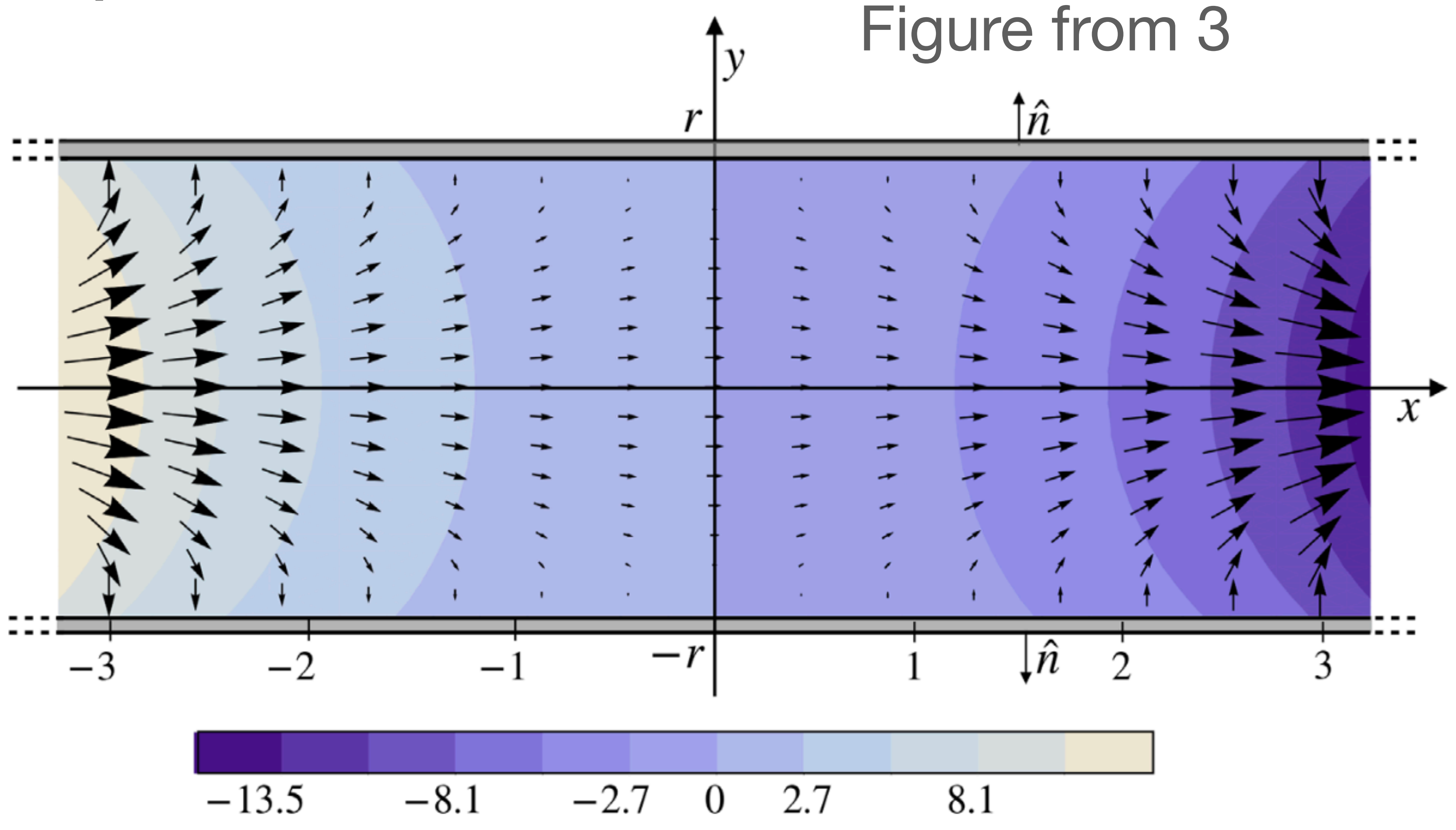


Problem 2: 2D Infinite Channel Flow

Model Validation and β calibration - Kristin Kurianski and Shilpa Khatri

- Utilizing problem and exact solution from Bernardi et al. (2023)[3]

Figure from 3



- Flow driven by pressure difference between interior and exterior

- Boundary condition from no-slip and Darcy's law

$$\mathbf{u} \cdot \hat{\tau} = 0 \text{ and } \mathbf{u} \cdot \hat{\mathbf{n}} = \mathrm{Da}\, p \text{ on } \Gamma$$

where Da is the Darcy number.

- Solution $\mathbf{u} = \left(-\frac{C}{\lambda_1}\cosh(\lambda_1 x) g_1'(y),\ C\sinh(\lambda_1 x) g_1(y)\right),\quad p = C\sinh(\lambda_1 x)\cos(\lambda_1 y),$

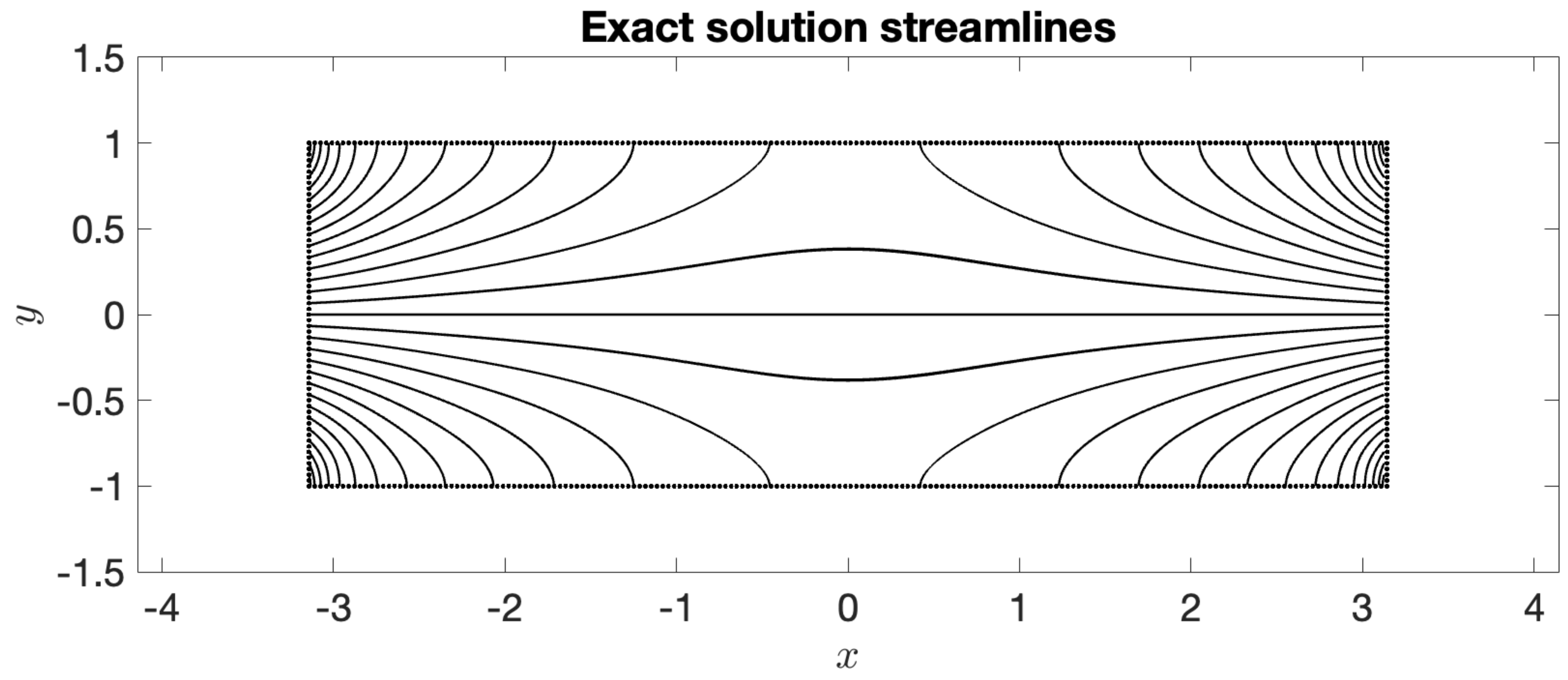
$$g_1(y) = \frac{\cos(\lambda_1)}{\sin(\lambda_1)}\left(\mathrm{Da} - \frac{1}{2}\right)\sin(\lambda_1 y) + \frac{1}{2} y\cos(\lambda_1 y) \text{ and } \lambda_1 = \text{root of } \tan^2(\lambda) - \frac{\tan(\lambda)}{\lambda} = 2\mathrm{Da} - 1$$

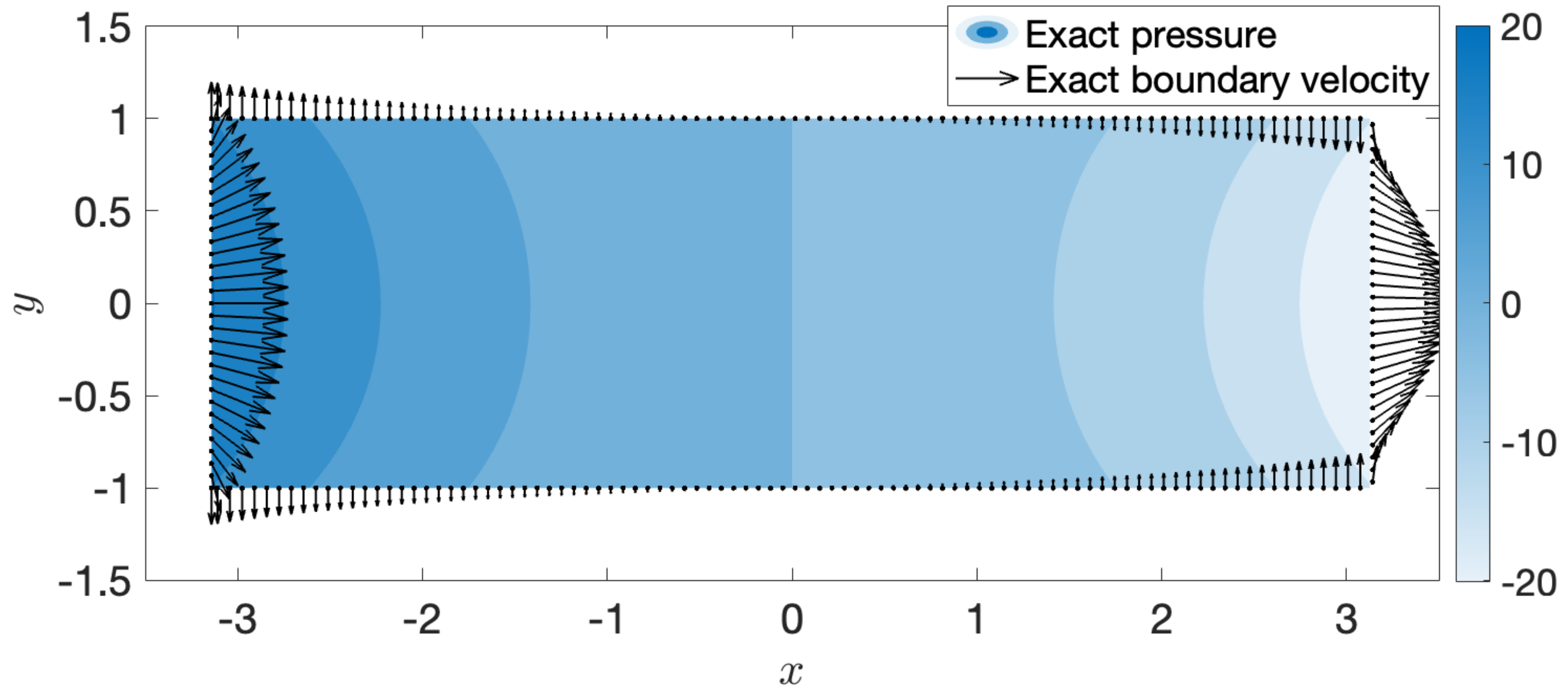
3. F. Bernardi, S. Chellam, N. Cogan, M. Moore Stokes flow solutions in infinite and semi-infinite porous channels, Studies in Applied Mathematics 151 (1) (2023) 116–140.

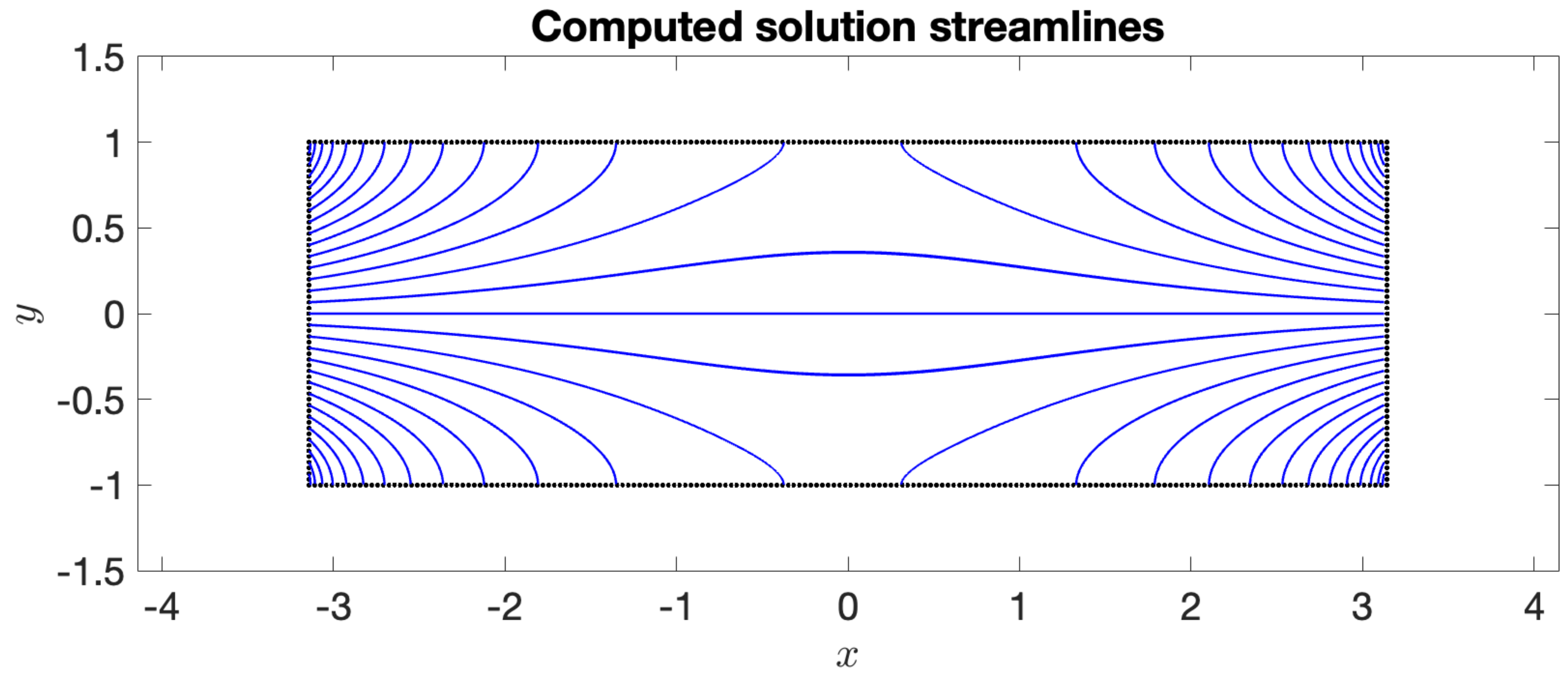
Problem 2: Preliminary Results

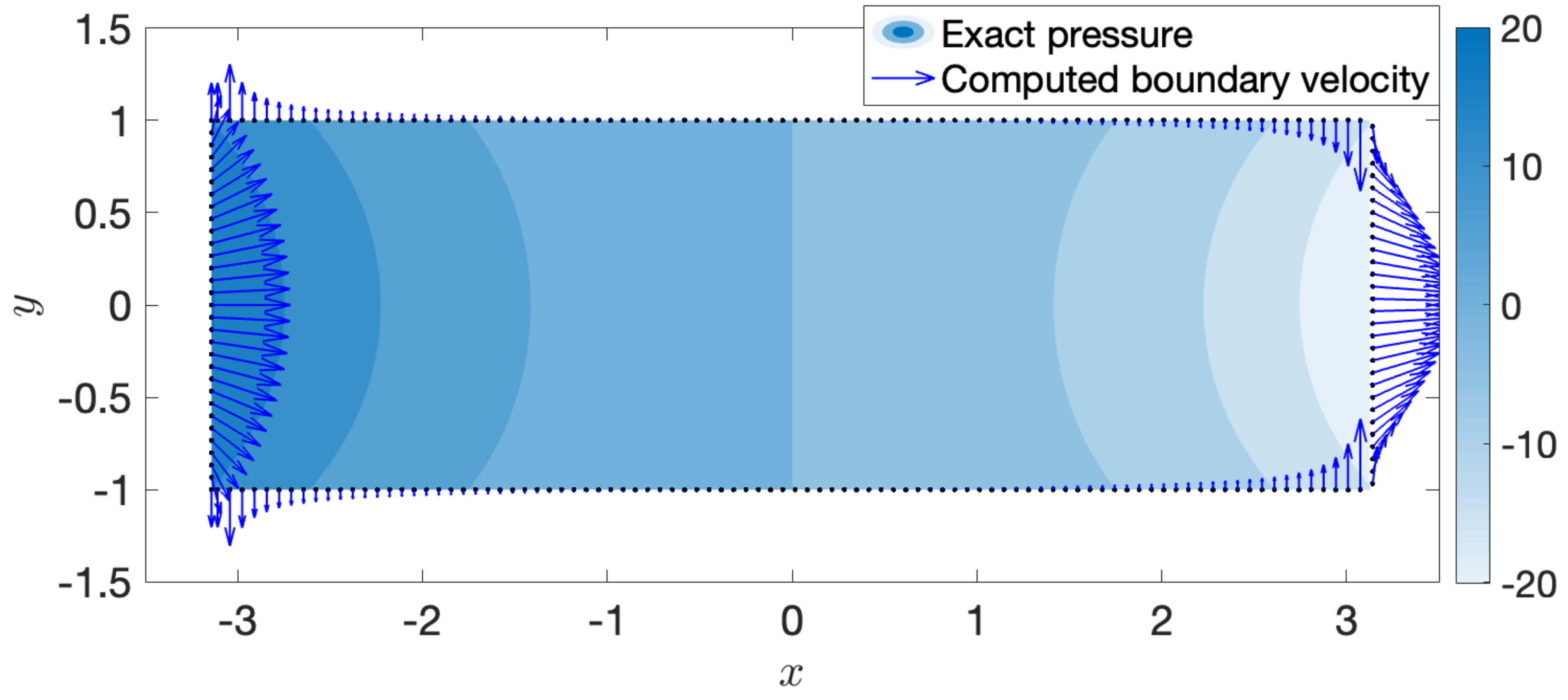
Preliminary Results

- Qualitative agreement in streamlines, have not yet achieved quantitative agreement
- Challenge: β is not constant but rather a function of x



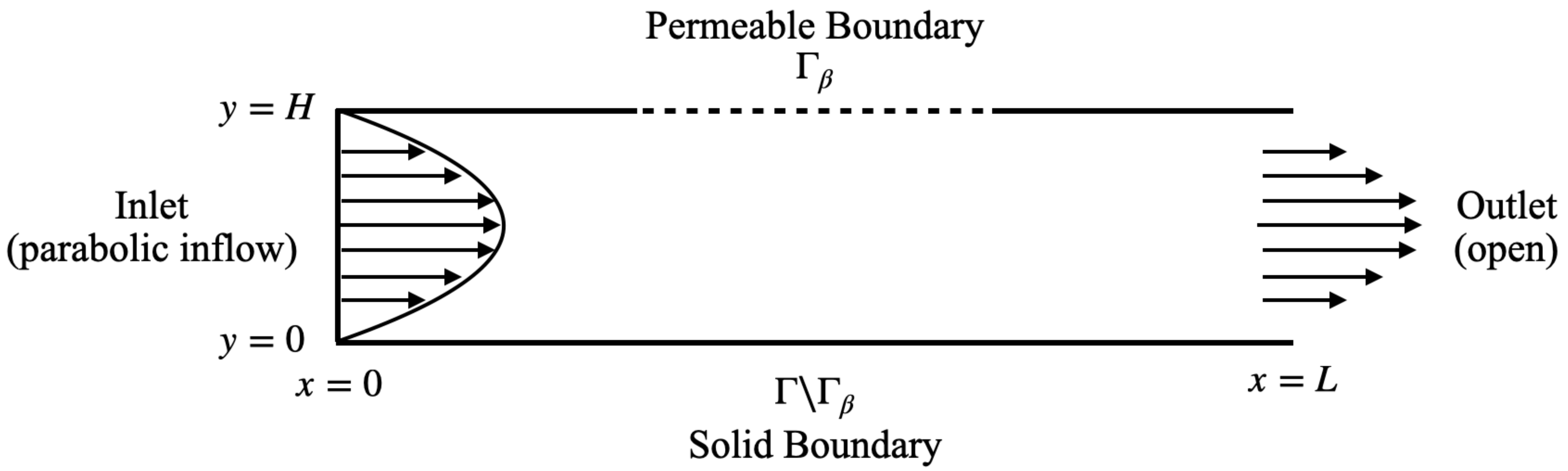


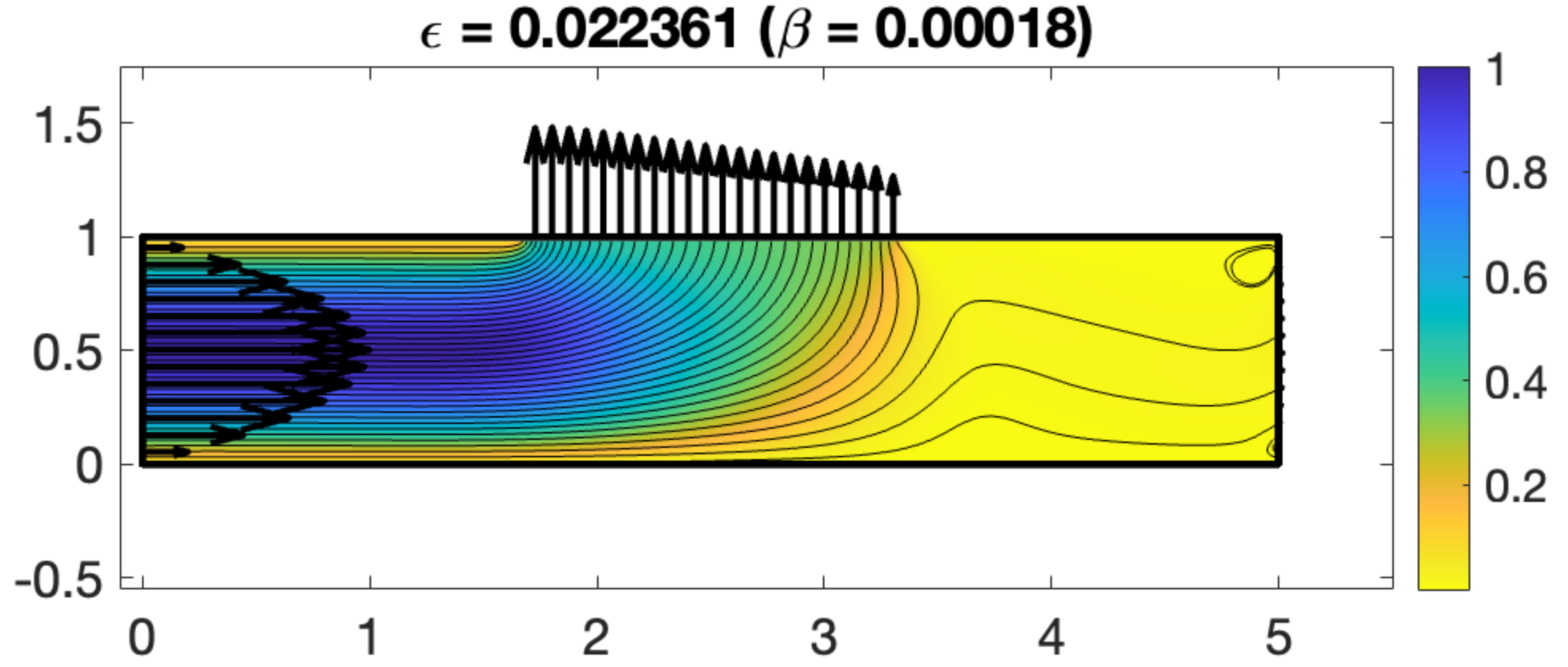




Problem 3: 2D Channel Flow

β Calibration, Convergence Analysis - Brittany Leathers, Michaela Kubacki



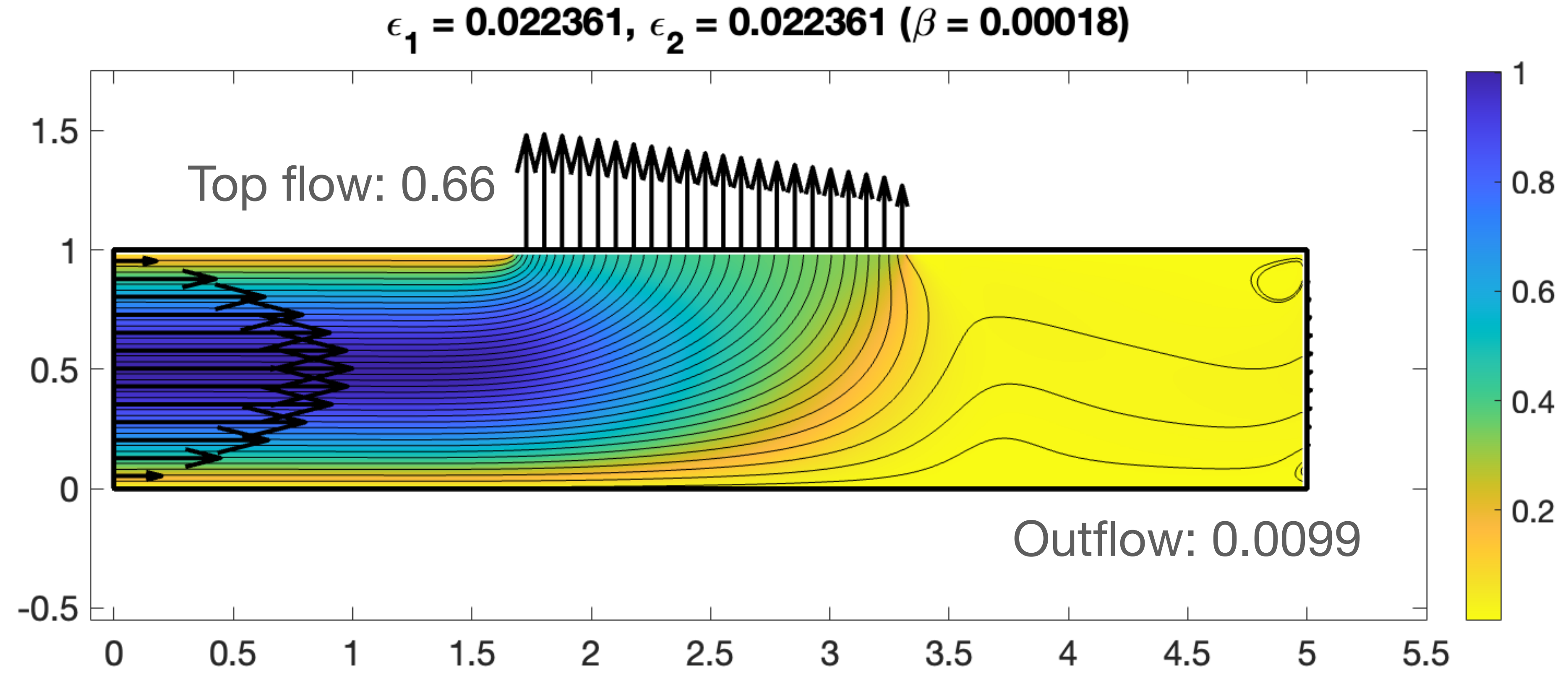


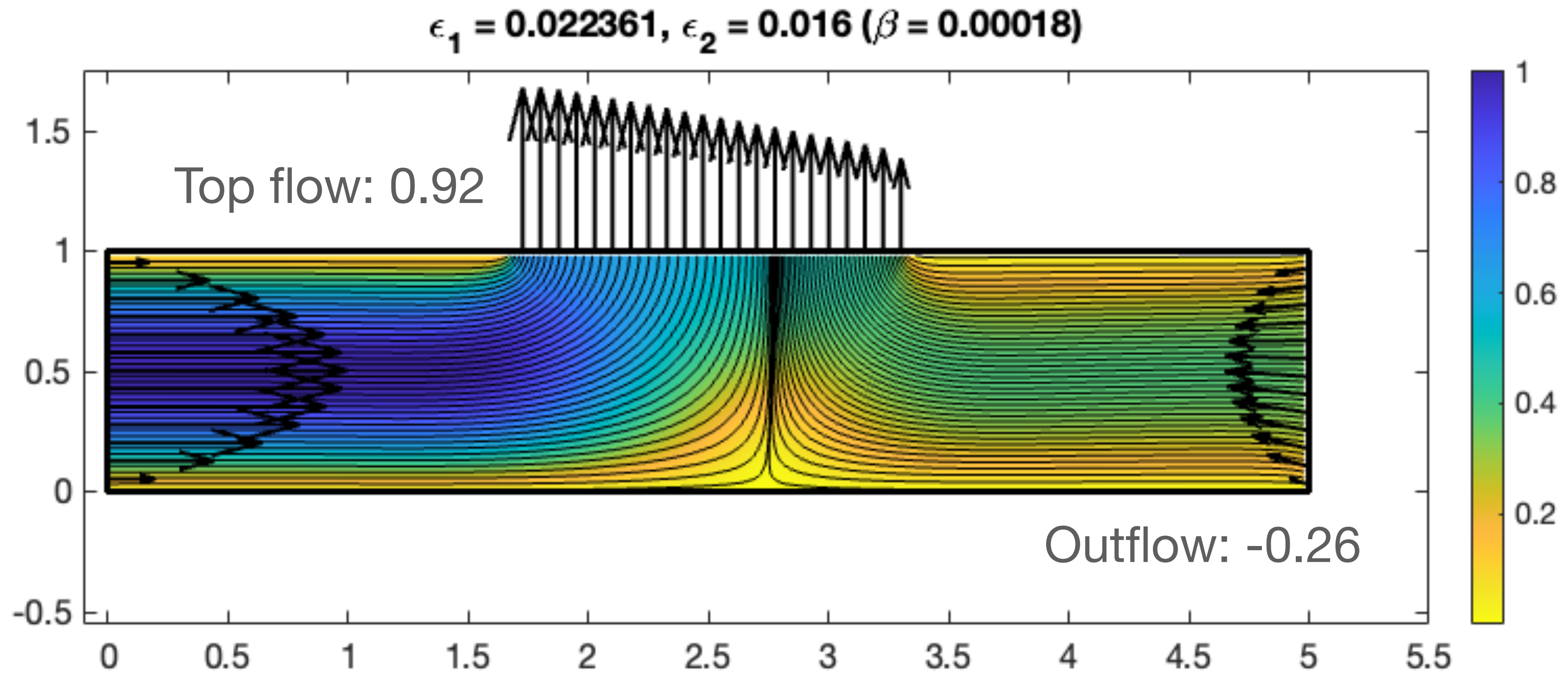
- Fluid velocity $\mathbf{u}_b$ on Γ_β is entirely due to source doublets (and unknown).
- Solving for velocity within the channel is a 4-step process:

1. Solve for $\mathbf{g}$ on Γ. $(\mathcal{M}_{St} + \mathcal{M}_{SD})\mathbf{g} = \mu \mathbf{u}_b$ on $\Gamma \backslash \Gamma_\beta$, $\quad \mathcal{M}_{St}\mathbf{g} = \mathbf{0}$ on Γ_β
2. Calculate the boundary velocity: $\mathcal{M}_{SD}\mathbf{g} = \mu \mathbf{u}_b$ on Γ_β
3. Solve for forces everywhere on channel boundary: $\mathcal{M}_{St}\mathbf{f} = \mu \mathbf{u}_b$ on Γ
4. Calculate channel velocity on entire domain: $\mathcal{M}_{St}\mathbf{f} = \mu \mathbf{u}$ in channel

Problem 3: Calibrating β Numerically

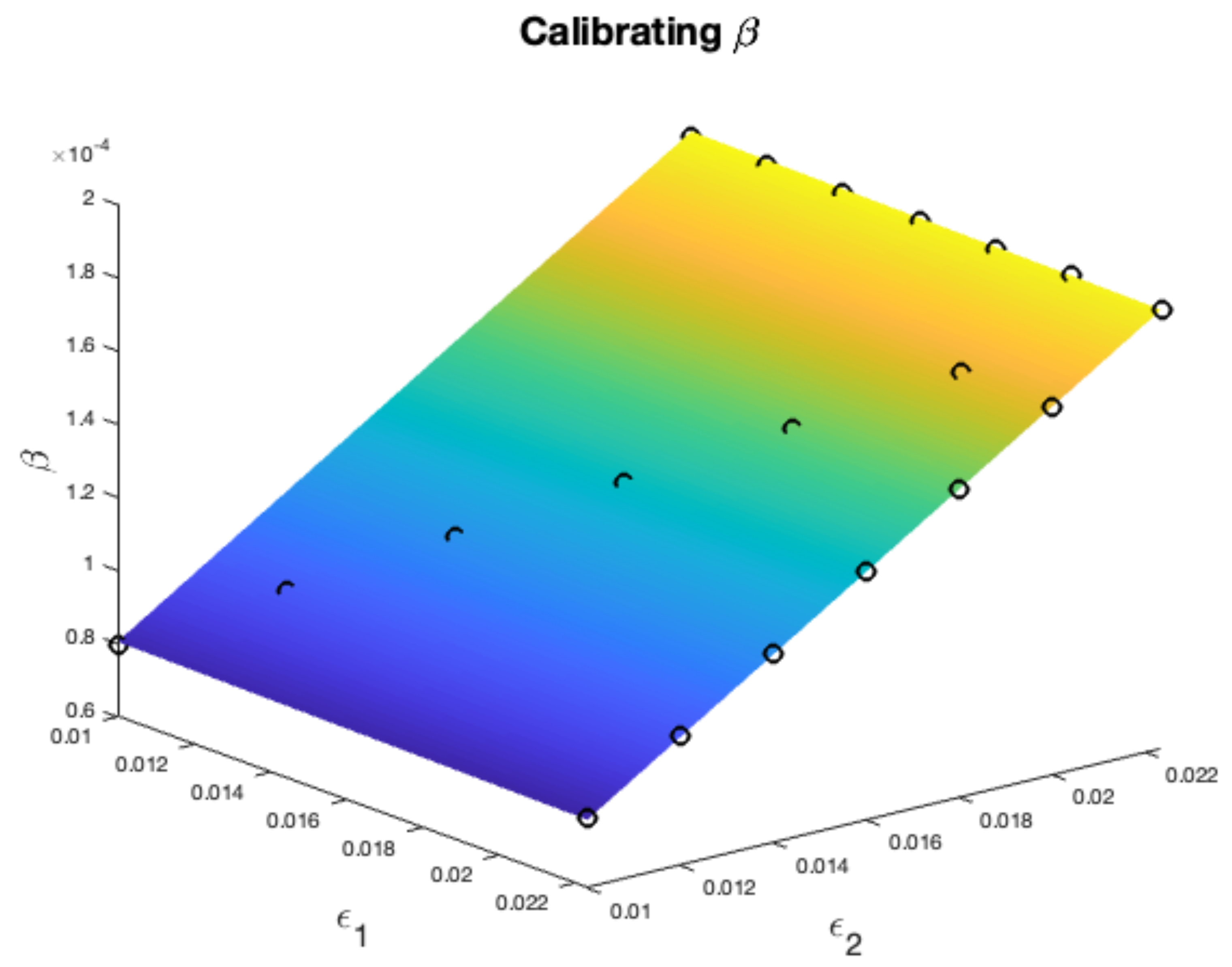
- β **depends on the blob, regularization size, and the permeability of the membrane!**





- **Numerical Experiment: Calibrate β**
  Select blob and set permeability (outflow).
  Numerically solve for the β value that matches the desired outflow. Change ϵ_1 and ϵ_2, and repeat.

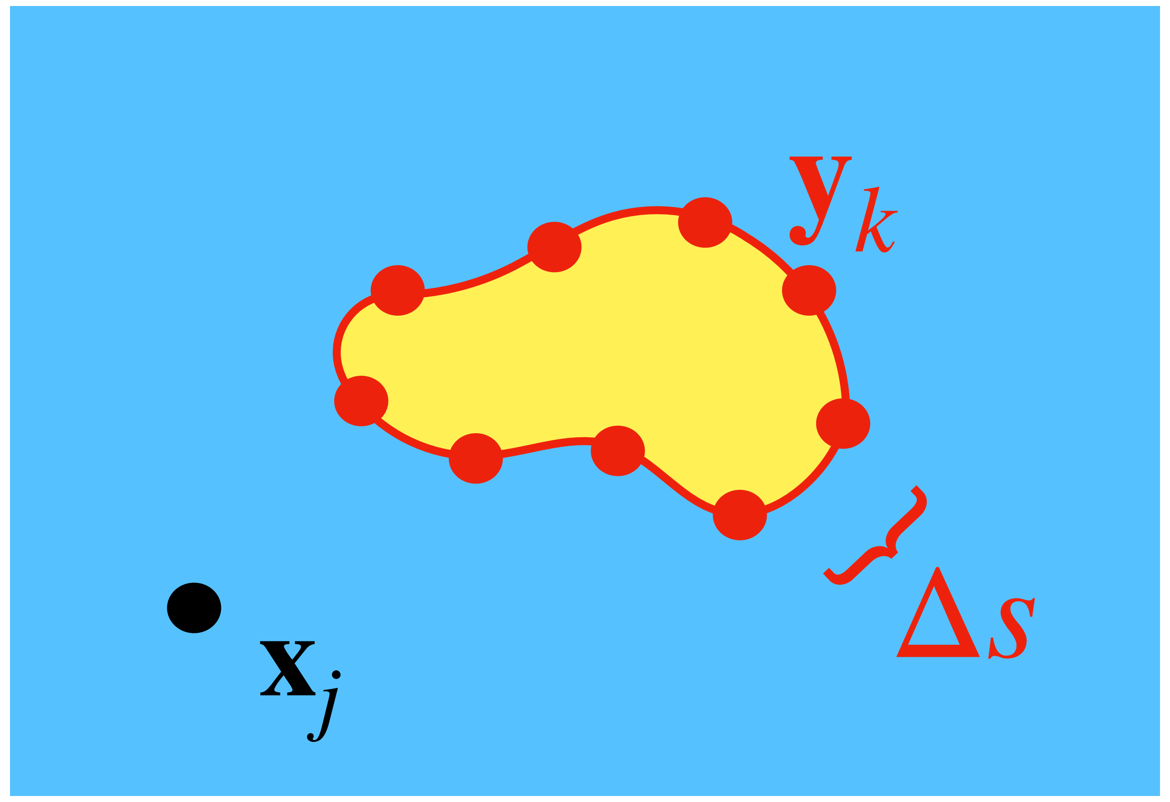
Result: $\beta = A\epsilon_1 + B\epsilon_2$



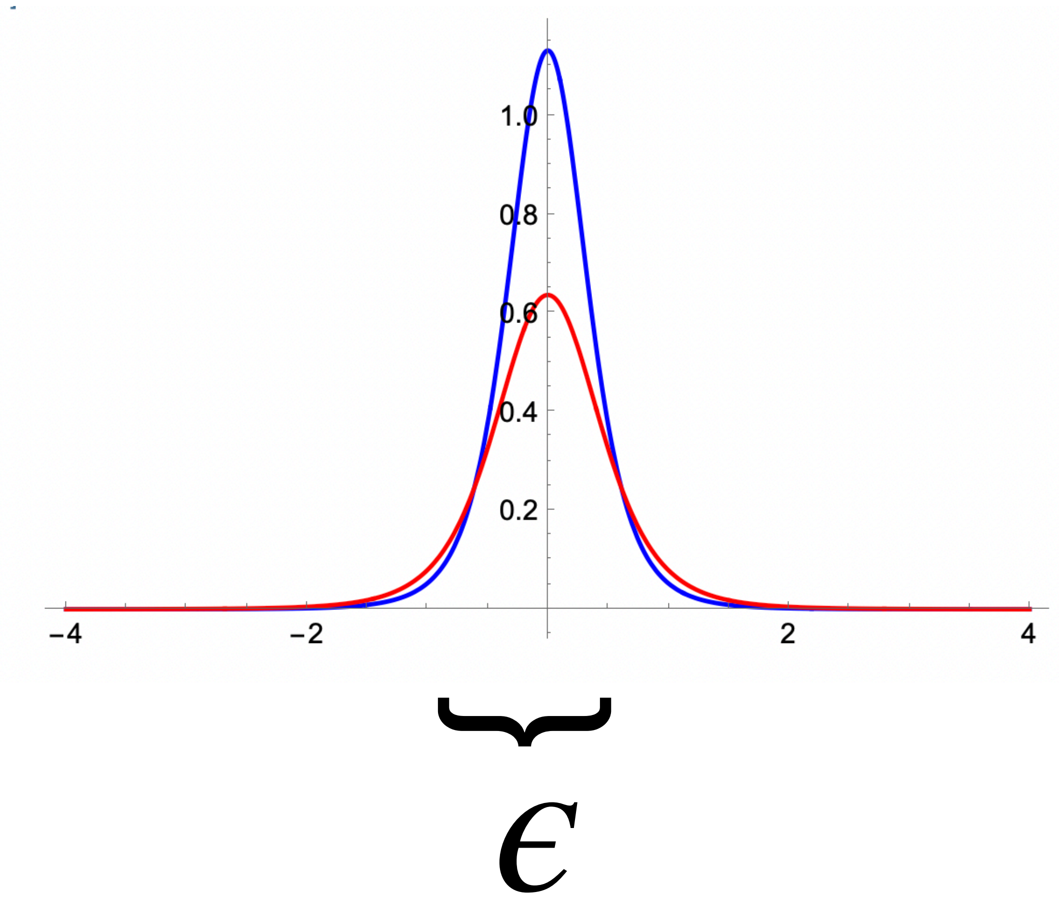
A little bit about error for MRS

Method of Regularized Stokeslets (MRS) Error

$$\begin{aligned}\text{Error} &= (\text{Regularization}) + (\text{Discretization})\\ &= E_{reg}(\phi_\epsilon, \epsilon) + E_{quad}(\Delta s, \epsilon)\\ &\approx C_1\epsilon^r + C_2\frac{\Delta s^p}{\epsilon^q}\end{aligned}$$



HOWEVER… in our case we can choose different regularization parameters for the **Stokeslets (ϵ_1)** and **Source Doublets (ϵ_2)**. This means our error is of form:



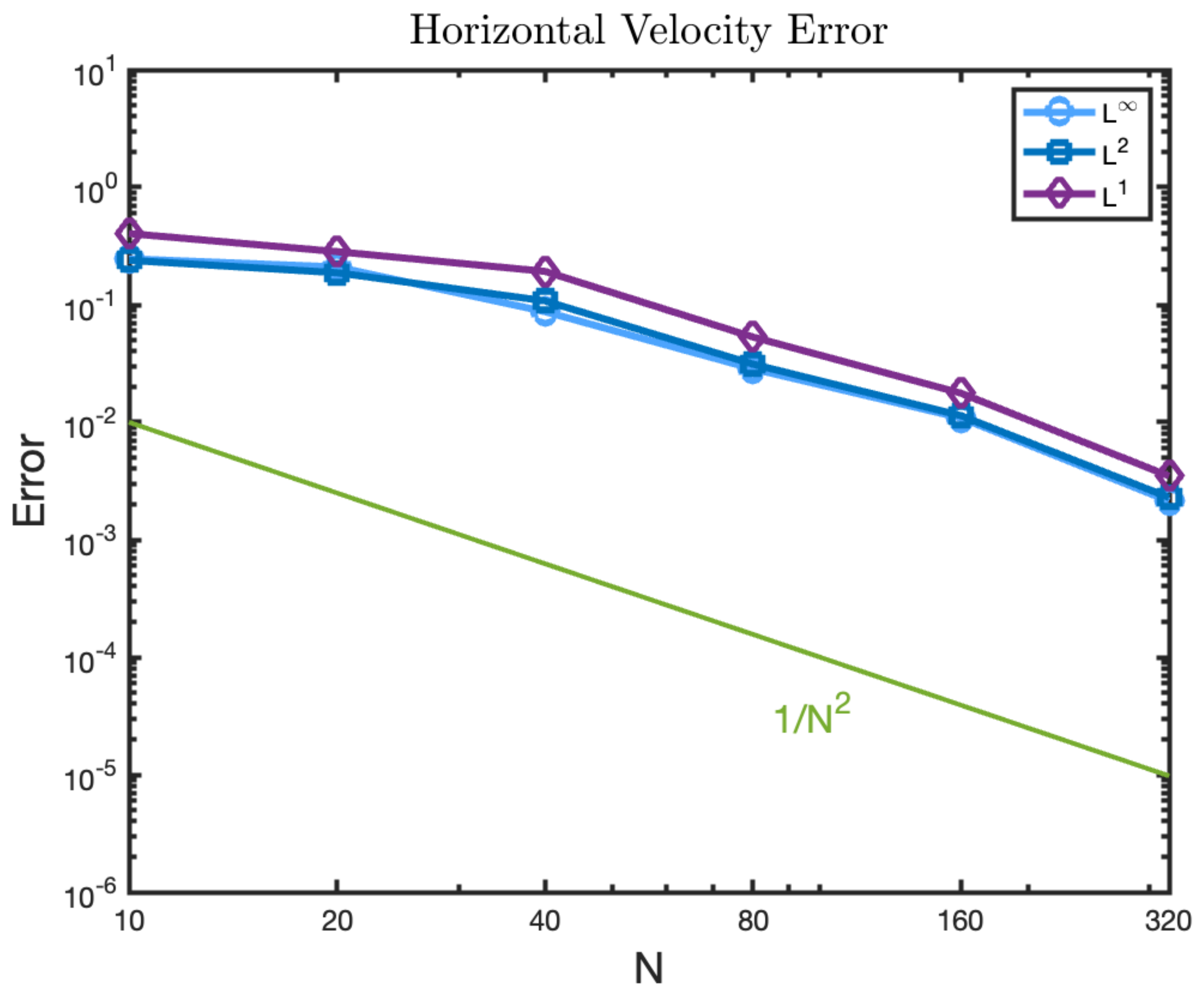
$$\begin{aligned}\text{Error} &= E_{Stokeslets} + E_{Source\ Doublets}\\ &\approx C_1\epsilon_1^{r_1} + C_2\epsilon_2^{r_2} + C_3\frac{\Delta s^{p_1}}{\epsilon_1^{q_1}} + C_4\frac{\Delta s^{p_2}}{\epsilon_2^{q_2}}\end{aligned}$$

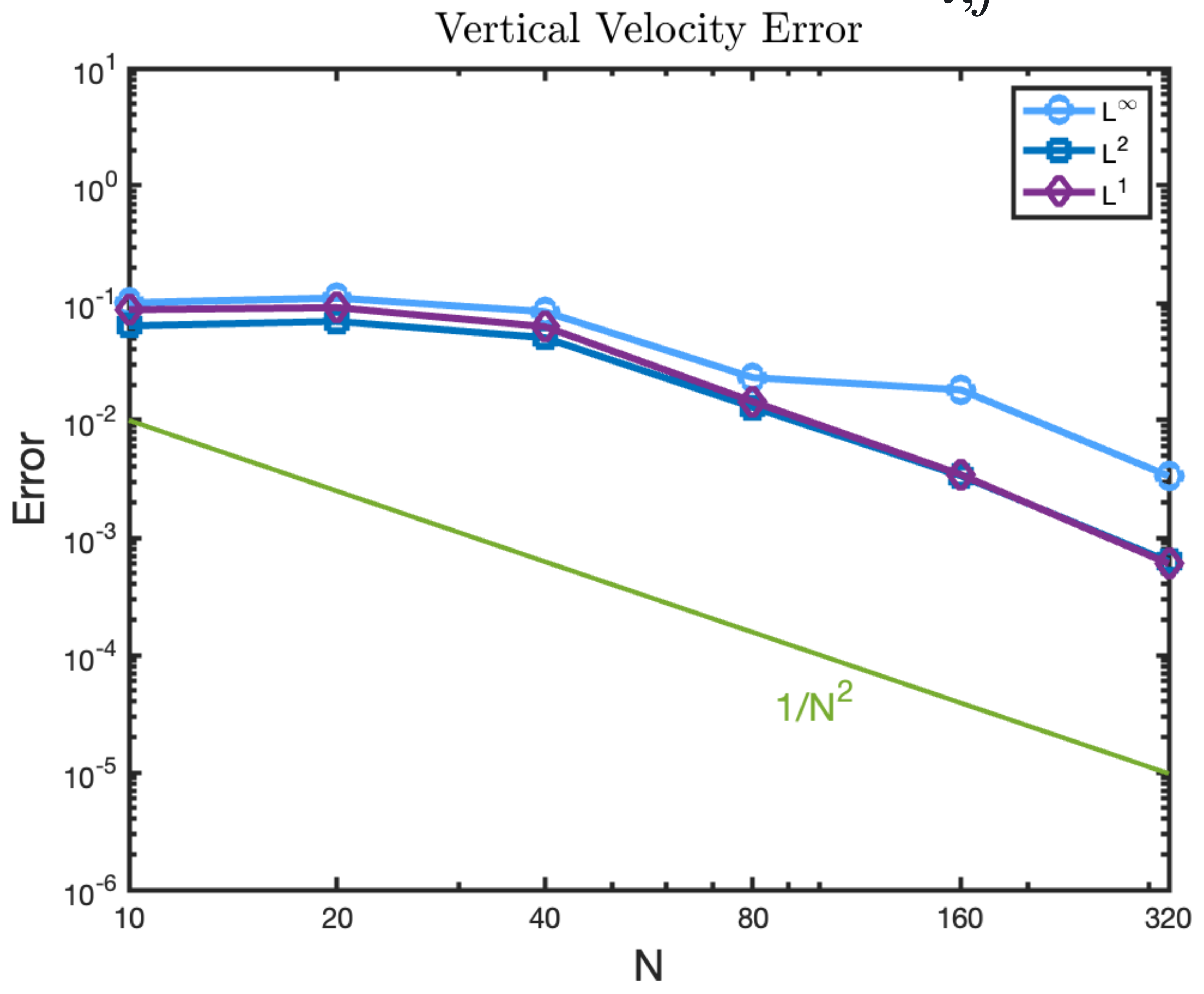
Problem 3: Prelimineary Convergence Studies

Preliminary Results

- Used $\epsilon_1 = C_1(ds)^{1/2}$ and $\epsilon_2 = C_2(ds)^{2/5}$, and adjusted β to preserve permeability
- Successive difference norms: $u = u_{coarse} - u_{fine}$

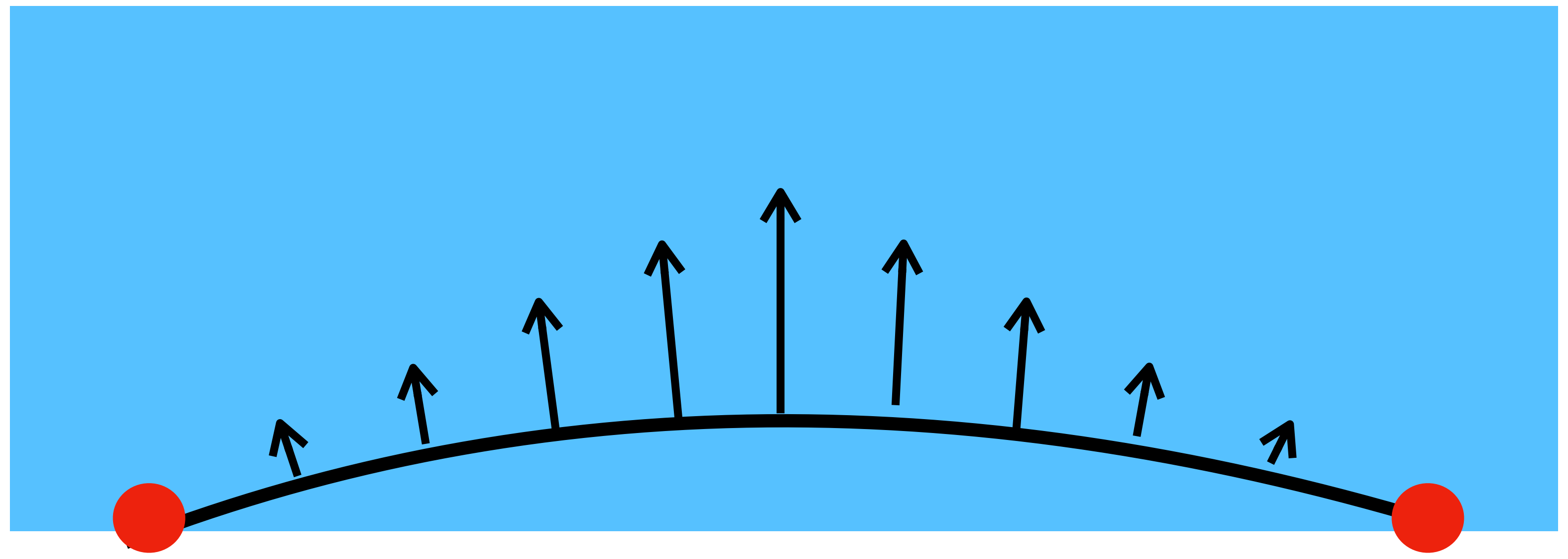
$$\|u\|_\infty = \max_{i,j} |u(x_i, y_j)|, \|u\|_2 = (\Delta x \Delta y \sum_{i,j} u(x_i, y_j)^2)^{1/2}, \|u\|_1 = \Delta x \Delta y \sum_{i,j} |u(x_i, y_i)|$$



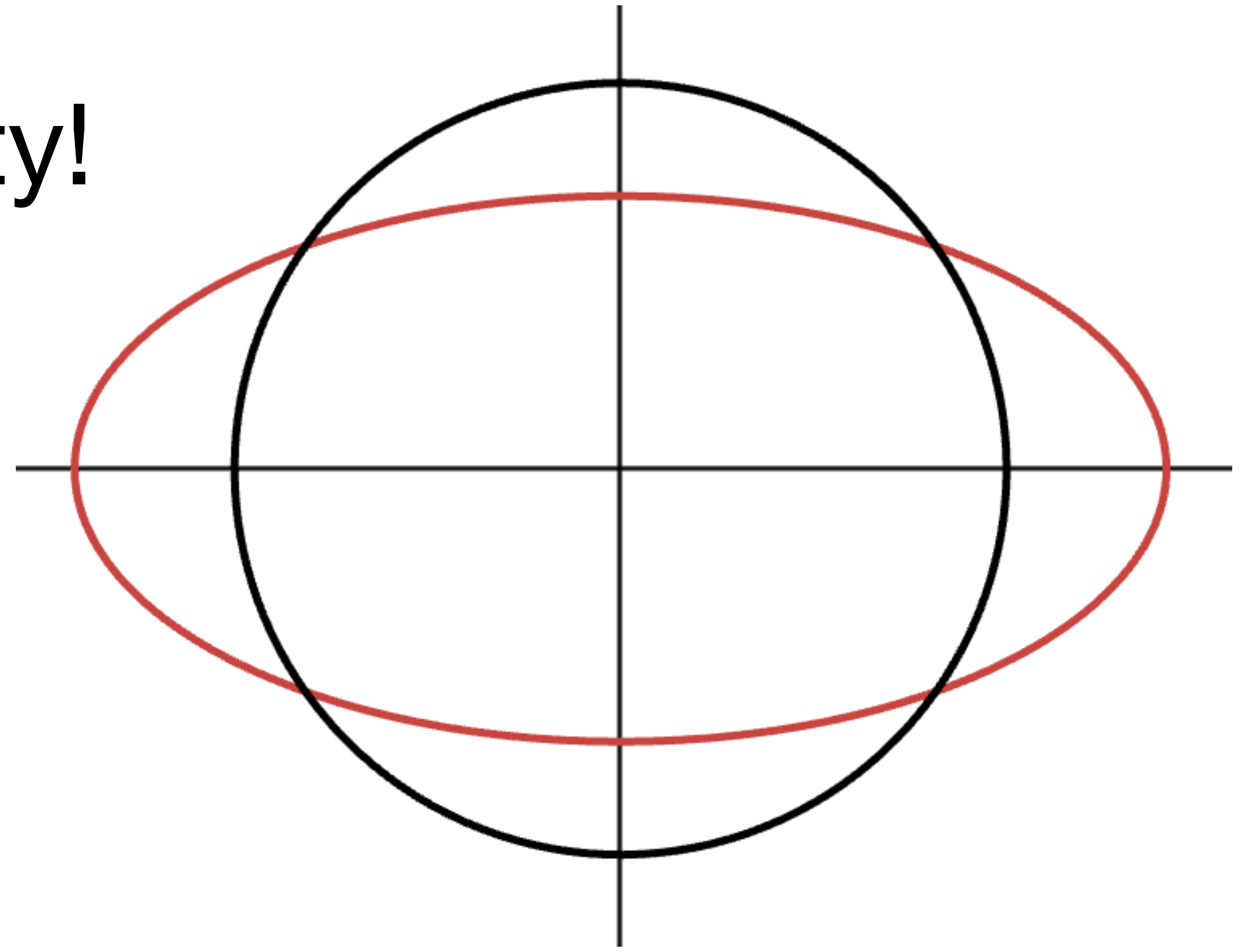


Extension: Volume Conservation

Source Doublet Correction - Ricardo Cortez and Terrance Pendleton



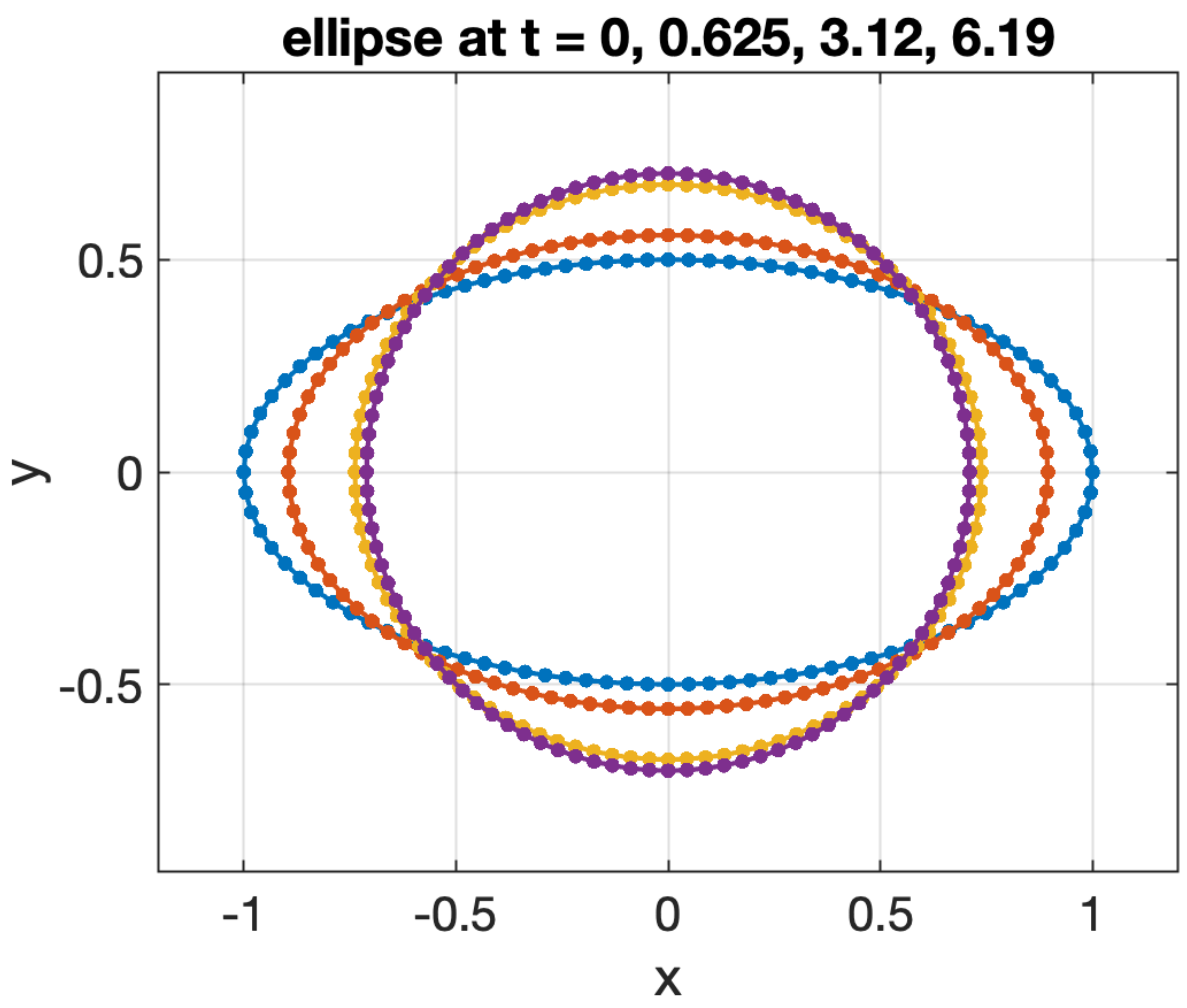
- Method of Regularized Stokeslets (MRS) can lead to **“Leaky Velocity”**
- **Example** Elastic ellipse relaxing to circle due to forces equal to curvature in normal direction.
    - Both the ellipse and the circle have the same area, but…
    - Solution computed using MRS suffers from leaky velocity!

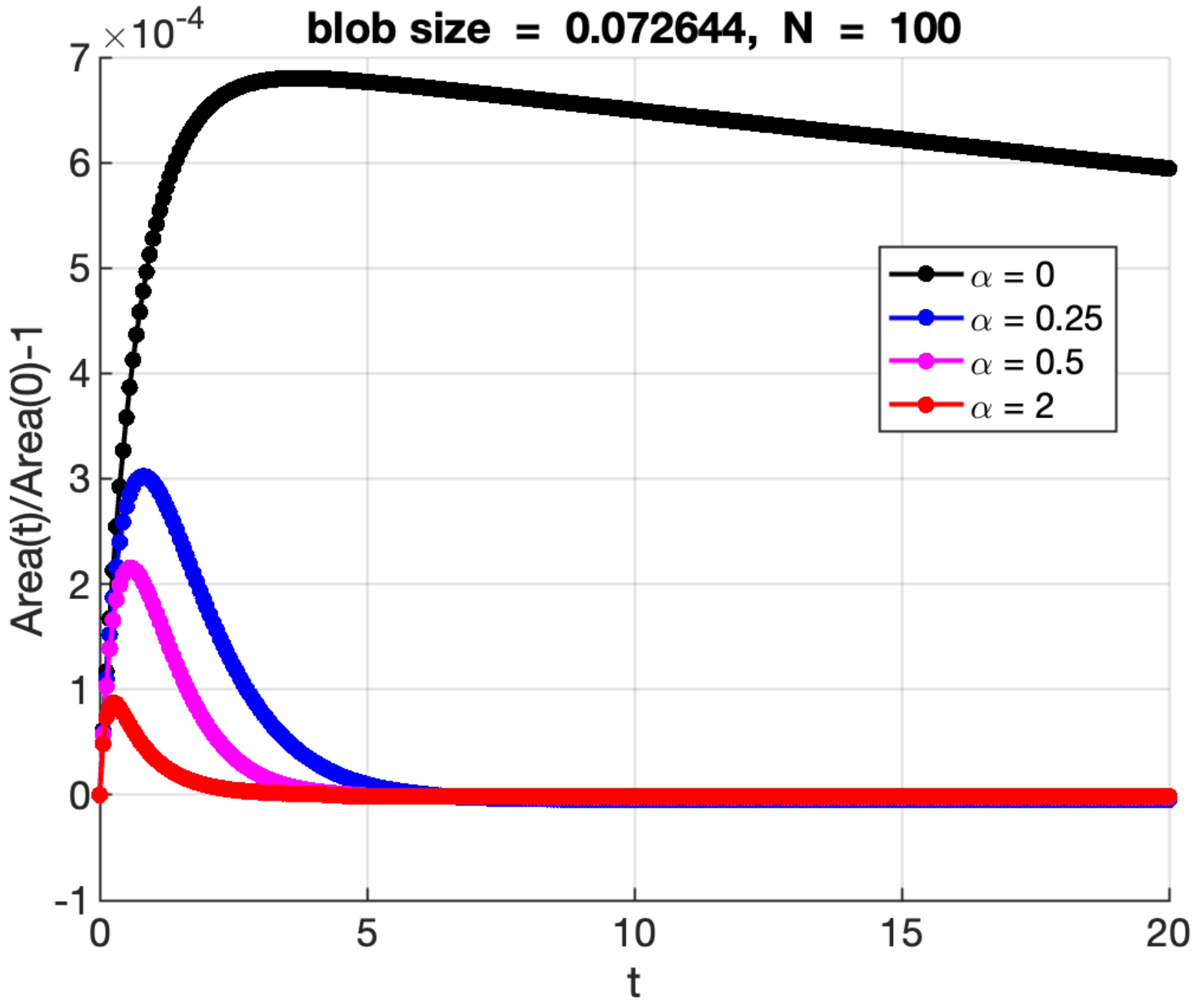


- **Idea:** Utilize source doublets! Impose source doublets at boundary source points as a corrective measure.

Volume Conservation Illustration

Preliminary Results

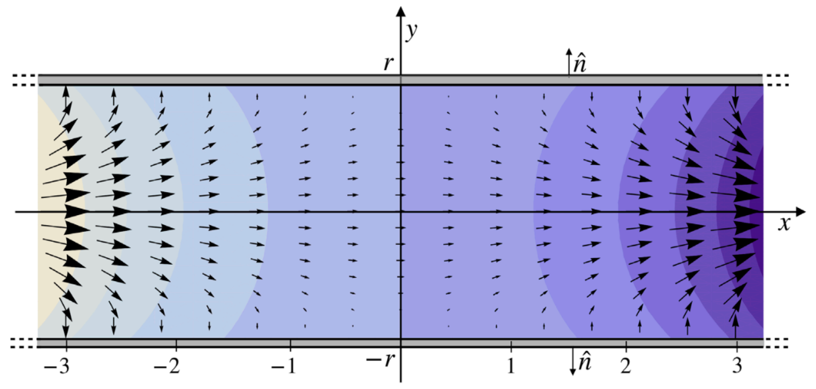




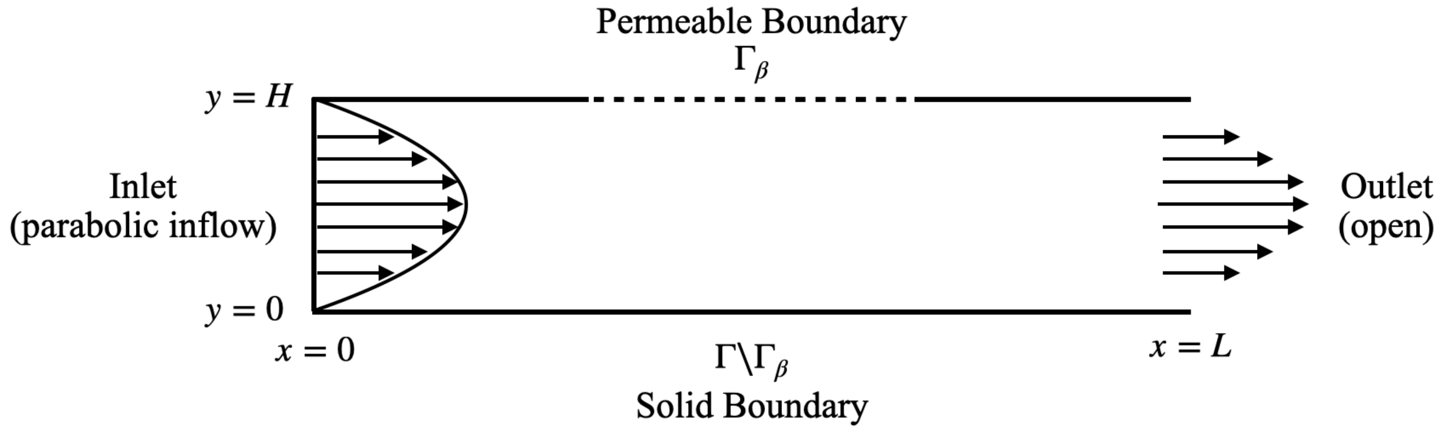
β is proportional to the area loss, α is the strength of the correction: $\beta = \alpha \left(\frac{\text{Area}(t)}{\text{Area}(0)} - 1 \right) \epsilon/(2\mu)$

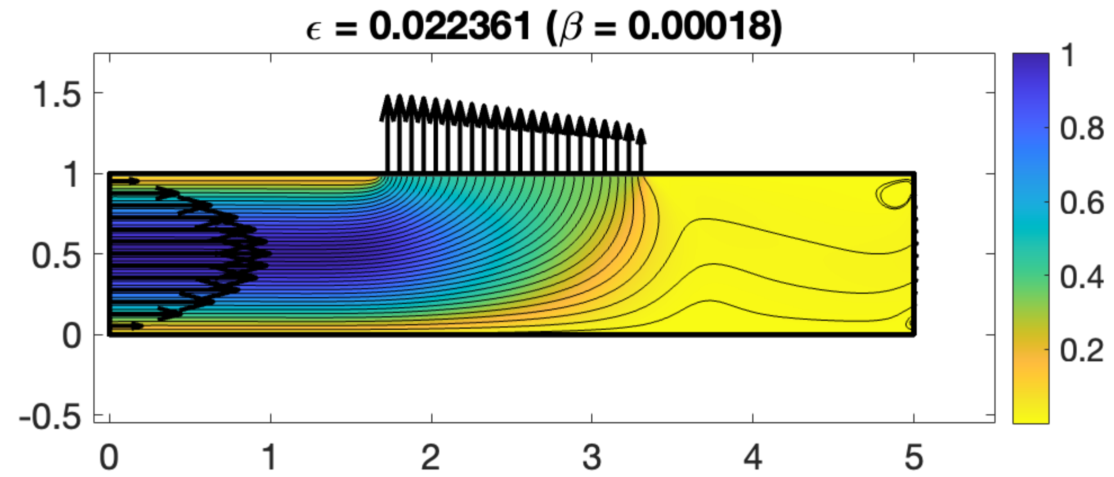
Ongoing and Future work

- Continue Model Validation and β calibration for 2D infinite and semi-infinite permeable channel flow using solutions from Bernardi et al. (2023)



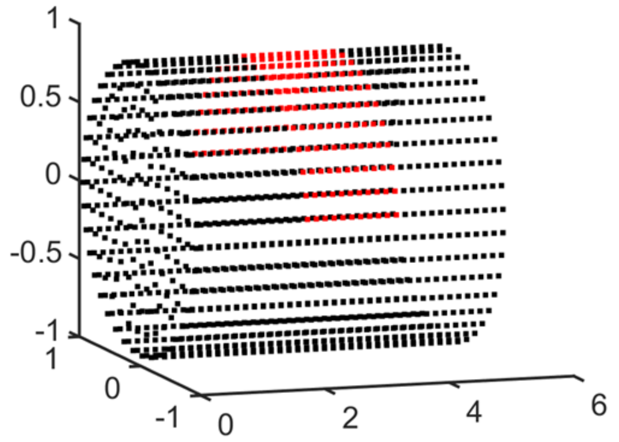
- Convergence analysis and optimal parameter selection for the 2D permeable Channel problem

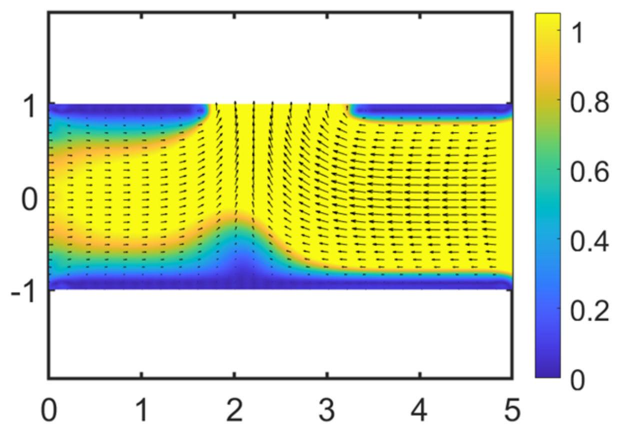


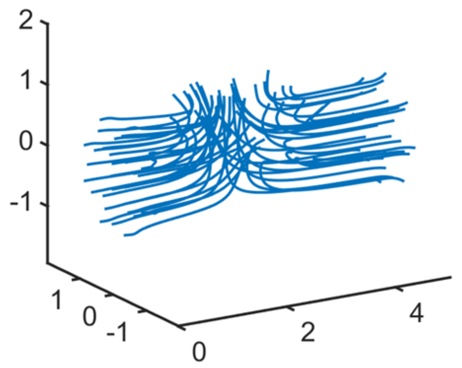


- Further exploration and application of using source doublets to mitigate volume loss

- Extension of the above to 3D permeable channels.







- Extension to biological applications.

Collaborate@ICERM

Summer 2026

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Solving the Point-Force Problem

1. Solve for ∇p and take divergence:

$$\nabla^2 p = \nabla \cdot \left(\mu \nabla^2 \mathbf{u} + \mathbf{f}\delta(\mathbf{x}-\mathbf{y})\right)$$

$$= \nabla \cdot (\mathbf{f}\delta(\mathbf{x}-\mathbf{y})) \qquad \text{since } \nabla \cdot \mathbf{u} = 0$$

$$= \nabla \cdot (\mathbf{f}\nabla^2 G(\mathbf{x}-\mathbf{y})) \qquad \text{since } \nabla^2 G = \delta$$

2. Solve for p: $p = \mathbf{f} \cdot \nabla G(\mathbf{x}-\mathbf{y})$
3. Plug in p to Stokes and split into 2 equations to solve for $\mathbf{u}$

$$\mu \nabla^2 \mathbf{u} = \nabla(\mathbf{f} \cdot \nabla G(\mathbf{x}-\mathbf{y})) - \mathbf{f}\delta(\mathbf{x}-\mathbf{y})$$

$$\mu \nabla^2 \mathbf{u}_1 = \nabla(\mathbf{f} \cdot \nabla G(\mathbf{x}-\mathbf{y}))$$

$$\mu \mathbf{u}_1 = (\mathbf{f} \cdot \nabla)\nabla B(\mathbf{x}-\mathbf{y})$$

$$\mu \nabla^2 \mathbf{u}_2 = -\mathbf{f}\delta(\mathbf{x}-\mathbf{y}) = -\mathbf{f}\nabla^2 G(\mathbf{x}-\mathbf{y})$$

$$\mu \mathbf{u}_2 = -\mathbf{f}G(\mathbf{x}-\mathbf{y})$$

$$\boxed{\mu \mathbf{u} = (\mathbf{f} \cdot \nabla)\nabla B(\mathbf{x}-\mathbf{y}) - \mathbf{f}G(\mathbf{x}-\mathbf{y})}$$

Stokeslet and Source Doublet Velocity Solutions

Choose a blob function ϕ_ϵ. Let $r = \|\mathbf{x} - \mathbf{X}(s)\|$. Our regularized velocity solutions have the following form.

Stokeslet Velocity: $$\mathbf{u}(\mathbf{x}) = \frac{1}{\mu}\int_\Gamma \mathbf{f}(s)H_1(r) + (\mathbf{f}(s)\cdot(\mathbf{x}-\mathbf{X}(s)))(\mathbf{x}-\mathbf{X}(s))H_2(r)\,ds$$

Source Doublet Velocity: $$\mathbf{v}(\mathbf{x}) = -\frac{1}{\mu}\beta(s)(\mathbf{f}(s)\cdot\hat{\mathbf{n}}(s))\left[\hat{\mathbf{n}}(s)S_1(r) + (\hat{\mathbf{n}}(s)\cdot\hat{\mathbf{n}})\hat{\mathbf{n}}S_2(r)\right]\,ds$$

where

$$H_1(r) = \frac{B'_{\epsilon_1}(r)}{r} - G_{\epsilon_1}(r) + \frac{1}{8\pi}, \quad H_2(r) = \frac{rB''_{\epsilon_1}(r) - B'_{\epsilon_1}(r)}{r^3}$$

$$S_1(r) = \frac{G'_{\epsilon_2}(r)}{r}, \quad S_2(r) = \frac{rG''_{\epsilon_2}(r) - G'_{\epsilon_2}(r)}{r^3}$$

Finding G_ϵ and B_ϵ: (2D)

1. Convert to $\nabla^2 G_\epsilon = \phi_\epsilon$ into polar form: $\dfrac{1}{r}\dfrac{\partial}{\partial r}(rG'(r)) = \phi_\epsilon(r)$

2. Solve for G'_ϵ: $G'_\epsilon(r) = \dfrac{\left(\int r\phi_\epsilon(r)\,dr + C_1\right)}{r}$

   Choose C_1 so that $\lim_{r\to 0} G'_\epsilon(r) = 0$

3. 3. Solve for G_ϵ: $G_\epsilon(r) = \displaystyle\int G'_\epsilon(r)\,dr + C_2$

   Choose C_2 so that $\lim_{r\to\infty} G_\epsilon = 0$